UNIT I — INTRODUCTION TO DATA SCIENCE

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1) Basics and Need of Data Science

What Data Science actually is

Data Science is not just “analyzing data.” That’s a weak definition.

It is:

A combination of statistics, programming, and domain knowledge used to extract useful, actionable insights from raw data.

Why it exists (Need)

Because raw data is useless until processed.

Real problem:

- Companies generate massive data
- Humans cannot manually analyze it
- Decisions must be data-driven, not intuition

Example

- Netflix recommending movies
- Amazon suggesting products

Key understanding

Without data science:

- Data = noise
  With data science:
- Data = decisions

---

2) Applications of Data Science

You need to understand categories, not random examples.

Major Applications

1. Healthcare

- Disease prediction
- Medical image analysis

2. Finance

- Fraud detection
- Risk analysis

3. E-commerce

- Recommendation systems
- Customer segmentation

4. Transportation

- Route optimization
- Self-driving cars

5. Social Media

- Sentiment analysis
- Content recommendation

Viva trap

If they ask: “Give application”
Don’t say just example → explain **how data is used**

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3) Relationship between Data Science, Information Science, Business Intelligence vs Data Science

This is where most students mess up.

Data Science vs Business Intelligence

Data Science	Business Intelligence
Future prediction	Past analysis
Uses ML	Uses dashboards
Complex models	Simple queries

Example

- BI: “Sales dropped last month”
- DS: “Sales will drop next month and why”

Data Science vs Information Science

Data Science	Information Science
Focus on insights	Focus on data organization
Uses algorithms	Uses storage/retrieval systems

Core Insight

- Information Science = managing data
- BI = reporting data
- Data Science = predicting using data

If you mix this up in viva, you’ll get grilled.

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4) Data Types

1. Structured Data

- Tabular (rows/columns)
- Example: Excel

2. Unstructured Data

- No fixed format
- Example: images, videos

3. Semi-structured Data

- Partial structure
- Example: JSON, XML

Critical Concept

Most real-world data is **unstructured**, not clean tables.

---

5) Data Lifecycle

This is very important.

Steps:

1. Data Collection
2. Data Storage
3. Data Processing
4. Data Analysis
5. Data Visualization
6. Decision Making

Reality

This is not linear. It is iterative.

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6) Methods: Data Cleaning, Integration, Reduction, Transformation, Discretization

This is the **core of Unit 1**.

(a) Data Cleaning

Removing garbage from data.

Problems:

- Missing values
- Noise
- Outliers

Techniques:

- Fill missing values
- Remove duplicates

(b) Data Integration

Combining multiple datasets.

Example:

- Sales data + Customer data

(c) Data Reduction

Reducing size but keeping information.

Methods:

- Sampling
- Feature selection

(d) Data Transformation

Changing format of data.

Examples:

- Normalization
- Scaling
- Encoding categorical values

(e) Data Discretization

Converting continuous data into categories.

Example:

- Age → Young / Adult / Old

---

7) Example: Academic Performance Dataset

From your syllabus

What you are expected to do

Step 1: Load data

```
df = pd.read_csv("students.csv")
```

Step 2: Check missing values

```
df.isnull().sum()
```

Step 3: Data cleaning

- Fill or remove missing values

Step 4: Transformation

- Normalize marks
- Encode categories

Step 5: Analysis

- Mean marks
- Pass/fail classification

What examiner actually checks

- Why you handled missing values that way
- Why normalization is needed

- Why not delete rows blindly

Mapping of Course Outcome (CO1)

This is not theory fluff. It means:

You should be able to:

- Understand raw data
- Clean it properly
- Prepare it for ML models

If you skip this, your models will fail silently.

Reality Check (Important)

If your understanding is:

- "Data cleaning removes null values"
- "Transformation means scaling"

Then you don't understand anything yet.

You need to know:

- When to remove vs when to fill
- When normalization is harmful
- When discretization destroys information

UNIT II — STATISTICAL INFERENCE

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1) Need of Statistics in Data Science

What statistics actually does

It lets you draw conclusions about a large population using a small sample.

Core problem

You never have full data. You always work with samples.

Example

- You don't check all customers
- You analyze a subset and generalize

Without statistics

- Your conclusions are guesses
- Your model outputs are meaningless

---

2) Measures of Central Tendency

These describe the "center" of data. But each one behaves differently.

Mean (Average)

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

Meaning

Sum of all values divided by total count.

Problem

Mean is **sensitive to outliers**

Example

Marks: 10, 20, 30, 1000
Mean = 265 → completely misleading

Median

Middle value after sorting.

Why it exists

To handle skewed data.

Example

Same data: 10, 20, 30, 1000
Median = 25 → more realistic

Mode

Most frequent value.

Use case

- Categorical data
- Market analysis (most common choice)

Key Insight

- Mean → mathematical center
- Median → robust center
- Mode → frequency center

If you mix these up, you don't understand data distribution.

---

3) Measures of Dispersion

These tell how spread out the data is.

Range

$$\text{Range} = \max(x) - \min(x)$$

Simple but useless for real analysis. One outlier breaks it.

Variance

$$\sigma^2 = \frac{1}{n}\sum\nolimits_{i=1}^{n}(x_i - \mu)^2$$

Measures how far values are from mean.

Problem

Units become squared → hard to interpret

Standard Deviation

$$\sigma = \sqrt{\frac{1}{n}\sum\nolimits_{i=1}^{n}(x_i - \mu)^2}$$

Fixes variance issue by taking square root.

Meaning

Average deviation from mean.

Mean Deviation

Average of absolute differences from mean.

Less used but conceptually simpler.

Key Understanding

- Low SD → data tightly clustered
- High SD → data widely spread

---

4) Bayes Theorem

This is not optional. This is core reasoning logic.

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$$

$$P(A)$$

$$P(B \mid A)$$

$$P(B \mid \neg A)$$

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)} \approx 0.68,\ P(B) \approx 0.25$$

P(B)=0.25P(B|A)P(A)=0.17P(A|B)~0.68Posterior = useful evidence / total evidence

---

Meaning

Probability of A given B.

Example

Disease testing:

- A = disease
- B = positive test

You want: P(disease | positive test)

Critical Insight

People confuse:

- $P(A|B) \neq P(B|A)$

That mistake destroys reasoning.

---

5) Basics and Need of Hypothesis Testing

What it is

A method to test assumptions using data.

Steps

1. Define Null Hypothesis (H_0)
2. Define Alternative Hypothesis (H_1)
3. Choose significance level (α)
4. Compute test statistic
5. Decide reject or not

Example

H_0: Average marks = 50
H_1: Average marks ≠ 50

Reality

You never "prove" something
You only **reject or fail to reject**

---

6) Pearson Correlation

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$$

Meaning

Measures linear relationship between two variables.

Range

- +1 → perfect positive
- 0 → no correlation
- -1 → perfect negative

Critical Mistake

Correlation ≠ causation

Example:

- Ice cream sales ↑ and drowning ↑
  Cause? Summer, not ice cream

---

7) Sample Hypothesis Testing

You test using sample, not population.

Why?

Population data is unavailable or expensive.

Types of Errors

Type I Error

Reject true hypothesis

Type II Error

Accept false hypothesis

Examiner trap

They will ask:
"Which is more dangerous?"

Answer:
Depends on context

- Medical → Type II is dangerous
- Law → Type I is dangerous

---

8) t-test

Used when:

- Small sample size
- Unknown population variance

Types

1. One-sample t-test
2. Independent t-test
3. Paired t-test

Core idea

Compare means using t-distribution.

Putting It Together (Example from syllabus)

Task:

- Calculate central tendency
- Calculate dispersion

Code Example

```
import pandas as pd

df = pd.read_csv("employee.csv")

mean = df['salary'].mean()
median = df['salary'].median()
mode = df['salary'].mode()[0]

variance = df['salary'].var()
std_dev = df['salary'].std()

print(mean, median, mode)
print(variance, std_dev)
```

What examiner checks

- Why median over mean?
- Why std dev matters?
- What if outliers exist?

Brutal Reality Check

If your answers sound like:

- "Variance measures spread"
- "Correlation shows relationship"

You will get destroyed in viva.

You need to explain:

- Why variance squares values
- Why correlation fails on nonlinear data
- Why hypothesis testing is probabilistic, not deterministic

UNIT III — DATA ANALYTICS LIFE CYCLE

Overview

This lifecycle defines how raw data becomes a usable solution.

The six phases in your syllabus:

1. Discovery
2. Data Preparation
3. Model Planning
4. Model Building
5. Communicate Results
6. Operationalize

Miss the purpose of even one phase and the entire pipeline collapses.

---

1) Discovery Phase

What actually happens

You define the **problem**, not touch data immediately.

Key tasks

- Understand business objective
- Identify success criteria
- Check data availability
- Define constraints

Example

Bad: “Build a prediction model”
Correct: “Predict student failure risk with ≥85% accuracy to enable intervention”

Questions you must answer

- What problem are we solving?
- Why does it matter?
- What data do we have?
- What is the output format?

Mistake

Jumping into coding without defining the problem.

That guarantees useless output.

---

2) Data Preparation Phase

This is the most time-consuming phase. Expect 60–80% effort here.

What happens

You convert raw, messy data into usable form.

Tasks involved

Data cleaning

- Handle missing values
- Remove duplicates
- Fix inconsistencies

Data transformation

- Normalization
- Encoding categorical variables

Feature engineering

- Create new variables
  Example:
  Date → extract day, month, year

Reality

If this phase is weak:

- Model accuracy will be fake
- Results will not generalize

---

3) Model Planning

What this means

You decide **how** to solve the problem.

Key decisions

- Classification or regression?
- Which algorithm?
- Evaluation metric?

Example

Problem: Predict pass/fail
→ Classification problem

Metrics selection

- Accuracy
- Precision
- Recall
- F1-score

Critical insight

Choosing wrong metric = wrong conclusion

Example:

- Fraud detection → accuracy is useless

---

4) Model Building

Now you actually train models.

Steps

1. Split dataset
    - Training set
    - Testing set
2. Train model
3. Tune parameters
4. Validate performance

Example (simple)

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression

X = df.drop('target', axis=1)
y = df['target']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

model = LogisticRegression()
model.fit(X_train, y_train)
```

Mistakes

- Training on full dataset → overfitting
- Ignoring validation

---

5) Communicate Results

Most students ignore this. That’s a mistake.

What this means

Explain results to **non-technical people**

Deliverables

- Visualizations
- Reports
- Insights

Example

Bad: “Model accuracy is 0.87”
Correct:
“87% of predictions are correct; main risk factor is attendance below 60%”

Reality

If stakeholders don’t understand:

- Your work is useless

---

6) Operationalize (Deployment)

This is where most academic projects stop. Real systems start here.

What happens

You put the model into real-world use.

Tasks

- Deploy model (API / app)
- Monitor performance
- Update model

Example

- Student prediction system integrated into college portal

Critical Insight

Model performance degrades over time → needs retraining

Case Study: GINA (Global Innovation Index Analysis)

From your syllabus.

What you are expected to understand

Step 1: Discovery

- Goal: Analyze innovation trends across countries

Step 2: Data Preparation

- Clean country-wise data
- Handle missing indicators

Step 3: Model Planning

- Choose analysis type (ranking, clustering)

Step 4: Model Building

- Apply statistical methods

Step 5: Communication

- Visual dashboards

Step 6: Operationalize

- Policy recommendation system

Mapping of CO3

You should be able to:

- Understand full analytics pipeline

- Build and deploy solution
- Interpret results correctly

Critical Fail Points

If you say:

- “Data preparation is cleaning data”
- “Model building is training model”

You’re giving shallow answers.

You must explain:

- Why feature engineering changes model output
- Why lifecycle is iterative
- Why deployment matters more than training

Reality Check

Most students:

- Focus only on model
- Ignore problem definition
- Ignore deployment

That’s why their projects are weak.

UNIT IV — PREDICTIVE DATA ANALYTICS WITH PYTHON

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1) Introduction

What “Predictive Analytics” actually means

You use historical data to predict unknown outcomes.

Types

- **Classification** → discrete output (pass/fail)
- **Regression** → continuous output (price, marks)

Reality

Prediction is probabilistic. You are not predicting truth, you are estimating likelihood.

---

2) Essential Python Libraries

Core libraries you must actually understand

- **pandas** → data manipulation
- **numpy** → numerical operations
- **matplotlib / seaborn** → visualization

- **sklearn** → ML models

What examiner checks

- Difference between .fit() and .predict()
- Why numpy is used internally in ML

If you can't answer that, you're just copying code.

---

3) Data Preprocessing

This is where most models succeed or fail.

(a) Removing Duplicates

df.drop_duplicates(inplace=True)

Why

Duplicate rows bias the model.

(b) Transformation

Scaling features to same range.

Types:

- Normalization (0–1)
- Standardization (mean = 0, std = 1)

(c) Data Fusion

Combining datasets from multiple sources.

(d) Mapping (Encoding)

Convert categorical → numerical

df['gender'] = df['gender'].map({'Male':0, 'Female':1})

(e) Replacing Missing Data

df.fillna(df.mean(), inplace=True)

Critical Insight

Bad preprocessing = misleading model accuracy

---

4) Analysis Types

Descriptive Analytics

What happened?

Example:

- Average marks

Predictive Analytics

What will happen?

Example:

- Will student pass?

Key Difference

Descriptive = past
Predictive = future

---

5) Classification vs Regression

Classification

Output is category

Examples:

- Spam / Not Spam
- Pass / Fail

Regression

Output is continuous

Examples:

- Salary prediction
- House price

Reality

Wrong choice here ruins everything.

---

6) Regression Algorithms

Linear Regression

$$y = \beta_0 + \beta_1 x$$

$$beta_0$$

$$beta_1$$

Meaning

Fits a straight line to data.

Assumption

Relationship is linear.

Problem

Fails completely on nonlinear data.

Example Code

```
from sklearn.linear_model import LinearRegression

model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

Examiner Trap

"What if relationship is not linear?"

Correct answer:
Model underfits → need nonlinear models

---

7) Logistic Regression

Despite the name, it is **classification**, not regression.

$$P(y = 1 \mid x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

Meaning

Outputs probability between 0 and 1.

Use case

Binary classification

Decision Boundary

If probability > 0.5 → class 1

Critical Mistake

Calling it regression because of name

---

8) Naive Bayes

Based on Bayes theorem.

Assumption

Features are independent

This assumption is almost always false, but model still works surprisingly well.

Use case

- Text classification
- Spam detection

Why it works

Probabilistic model, fast and scalable

---

9) Decision Trees

Concept

Split data based on conditions.

Example:

if marks > 50 → pass
else → fail

Advantages

- Easy to understand
- Handles nonlinear data

Problems

- Overfitting
- Sensitive to small data changes

---

10) Scikit-learn Introduction

What it is

Python library for ML models.

Basic Workflow

```
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier

X_train, X_test, y_train, y_test = train_test_split(X, y)

model = DecisionTreeClassifier()
model.fit(X_train, y_train)
pred = model.predict(X_test)
```

Key Functions

- .fit() → train model

- .predict() → make predictions
- .score() → evaluate

---

11) Dataset: IRIS Dataset

What it is

Classic dataset for classification.

Features:

- Sepal length
- Sepal width
- Petal length
- Petal width

Target:

- Species

Why used

- Clean
- Small
- Ideal for beginners

---

12) Filling Missing Values

Already covered but important here.

Methods

- Mean
- Median
- Mode

When to use

- Mean → normal distribution
- Median → skewed data

---

13) Regression and Classification using Scikit-learn

Regression Example

```
from sklearn.linear_model import LinearRegression
model = LinearRegression()
model.fit(X, y)
```

Classification Example

```
from sklearn.tree import DecisionTreeClassifier
model = DecisionTreeClassifier()
model.fit(X, y)
```

Example Task from Syllabus

Use IRIS dataset and apply preprocessing.

Full Flow

```
from sklearn.datasets import load_iris
import pandas as pd

data = load_iris()
df = pd.DataFrame(data.data, columns=data.feature_names)

df['target'] = data.target

# Check missing values
print(df.isnull().sum())

# Train model
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier

X = df.drop('target', axis=1)
y = df['target']

X_train, X_test, y_train, y_test = train_test_split(X, y)

model = DecisionTreeClassifier()
model.fit(X_train, y_train)

print(model.score(X_test, y_test))
```

Critical Fail Points

If you say:

- “Logistic regression is used for regression”
- “Naive Bayes assumes real-world independence”

You don’t understand anything.

UNIT V — DATA ANALYTICS AND MODEL EVALUATION

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1) Clustering Algorithms

Clustering is **unsupervised learning**. No labels. You group data based on similarity.

K-Means Clustering

Core idea:
Split data into **K groups** where each point belongs to the nearest cluster center.

Objective function

$$J = \sum_{i=1}^{k} \sum_{x \in C_i} \| x - \mu_i \|^2$$

Steps

1. Choose K
2. Initialize centroids
3. Assign points to nearest centroid
4. Update centroids
5. Repeat until convergence

Critical problems

- You must choose **K manually**
- Sensitive to initialization
- Assumes spherical clusters

Examiner trap

"What if clusters are not circular?"

Correct answer:
K-Means fails → use DBSCAN or hierarchical clustering

---

2) Hierarchical Clustering

Builds a tree (dendrogram) of clusters.

Types

- Agglomerative (bottom-up)
- Divisive (top-down)

Advantage

No need to predefine K

Problem

- Computationally expensive
- Hard to scale

---

3) Time Series Analysis

Used when data depends on time.

Examples

- Stock prices
- Temperature data

Key components

- Trend
- Seasonality
- Noise

Reality

Random patterns often look meaningful. Most students misinterpret noise as trend.

---

4) Introduction to Text Analytics

Text is unstructured → needs conversion.

Steps

1. Tokenization
2. Stopword removal
3. Stemming / Lemmatization

Problem

Text loses meaning during preprocessing.

---

5) Bag of Words (BoW)

Convert text → numerical vectors.

Idea

Count word frequency.

Example:
"data science is fun"
→ {data:1, science:1, fun:1}

Problem

- Ignores context
- Treats all words equally

---

6) TF-IDF

Improves BoW by reducing importance of common words.

Formula

$$TF\text{-}IDF = TF \times \log\left(\frac{N}{DF}\right)$$

Meaning

- TF → frequency in document
- IDF → rarity across documents

Insight

Rare words carry more importance

---

7) Evaluation of Classification Models

This is where students fail badly.

Confusion Matrix

$$\begin{bmatrix} TP & FP \\ FN & TN \end{bmatrix}$$

Terms

- TP → correct positive
- TN → correct negative
- FP → false positive
- FN → false negative

---

8) Metrics

Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Problem

Fails on imbalanced data

Example:
99% negative → model predicts always negative → 99% accuracy → useless

---

Precision

$$\text{Precision} = \frac{TP}{TP + FP}$$

Meaning

How many predicted positives are correct

Recall

$$\text{Recall} = \frac{TP}{TP + FN}$$

Meaning

How many actual positives are detected

F1 Score

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

Why it exists

Balances precision and recall

---

9) Model Evaluation Techniques

Cross Validation

Split data multiple times.

Types

- K-Fold
- Stratified

Why needed

Single train-test split is unreliable

---

10) ROC Curve

Plots:

- True Positive Rate (Recall)
- False Positive Rate

Insight

Shows model performance across thresholds

AUC (Area Under Curve)

- 1 → perfect
- 0.5 → random

---

11) Parameter Tuning

Adjust model parameters for better performance.

Example

- Depth of decision tree
- Learning rate

Method

- Grid Search
- Random Search

---

12) Metrics for Clustering

Elbow Method

Find optimal K

Idea

Plot:

- K vs error

Choose point where improvement slows

Problem

Elbow is often unclear

Example Task from Syllabus

Use IRIS dataset and apply K-Means.

Code

```
from sklearn.datasets import load_iris
from sklearn.cluster import KMeans

data = load_iris()
X = data.data

model = KMeans(n_clusters=3)
model.fit(X)

print(model.labels_)
```

Critical Fail Points

If you say:

- “Accuracy is best metric”

- “K-Means always works”

You don’t understand evaluation.

UNIT VI — DATA VISUALIZATION AND HADOOP

PART 1: DATA VISUALIZATION

---

1) Introduction to Data Visualization

What it actually is

Transforming data into visual form so patterns, trends, and anomalies become obvious.

Why it exists

Humans cannot interpret large tables efficiently. Visuals compress information.

Reality

Bad visualization = wrong conclusions
Good visualization = faster and correct decisions

---

2) Importance of Visualization

Without visualization

- Data is unreadable
- Insights are hidden

With visualization

- Patterns become visible
- Outliers are detected
- Trends are understood

Example

Raw data:

Marks: 45, 50, 48, 47, 95

Table won’t show issue clearly
Plot will expose outlier (95)

---

3) Principles of Good Visualization

Clarity

Graph must be easy to understand

Accuracy

Do not distort data

Example:
Truncated Y-axis → misleading growth

Efficiency

Deliver maximum information with minimum clutter

Consistency

Same color = same meaning across charts

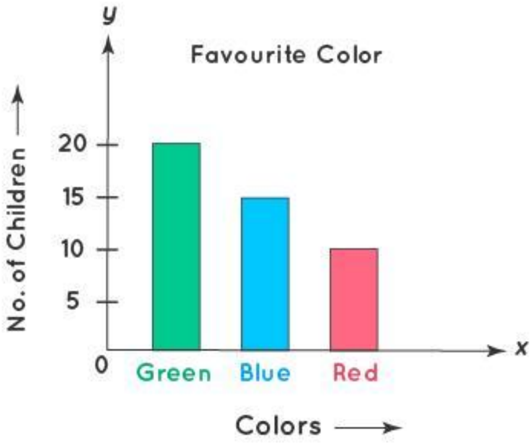
Critical failure

Over-decorated graphs with no insight

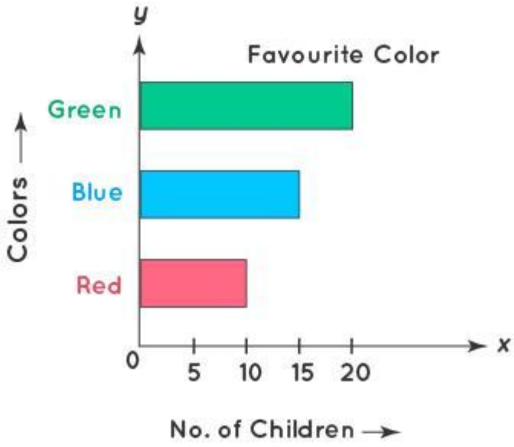
---

4) Types of Charts

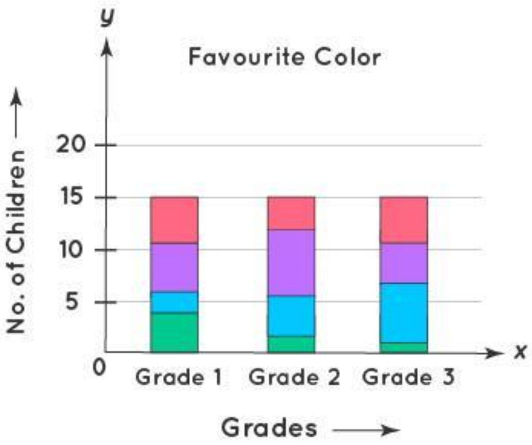
Bar Chart



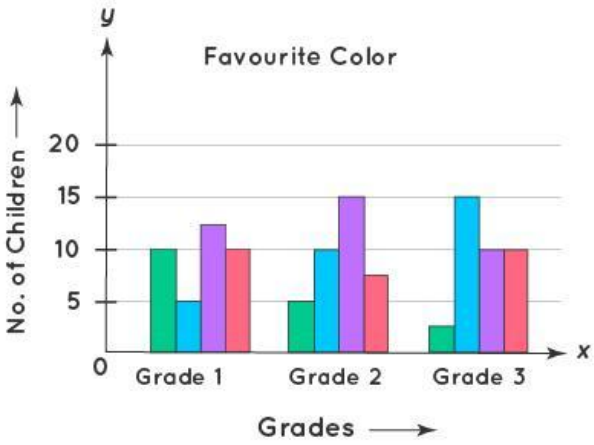
Vertical Bar Graph



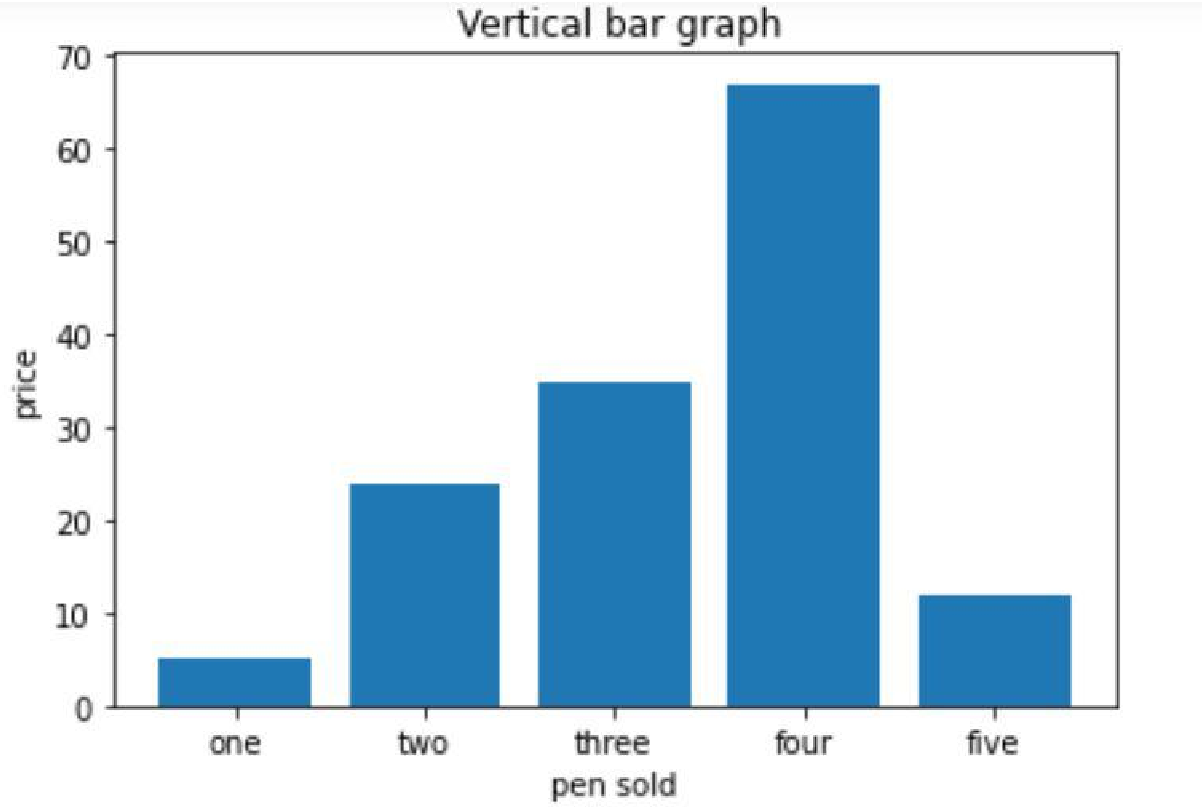
Horizontal Bar Graph

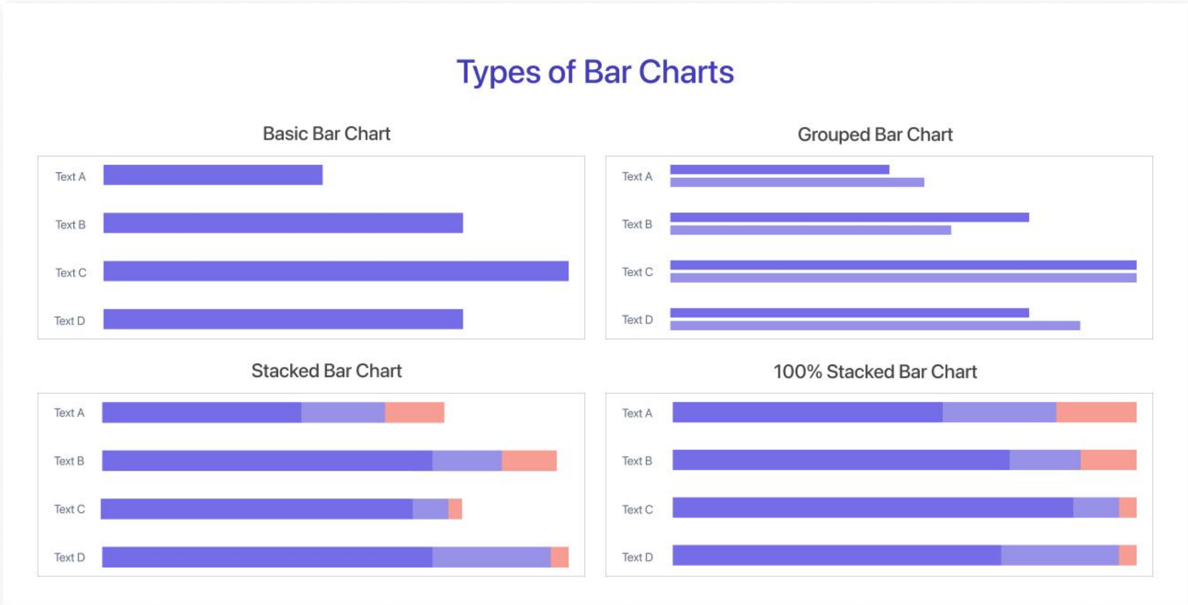


Stacked Bar Graph

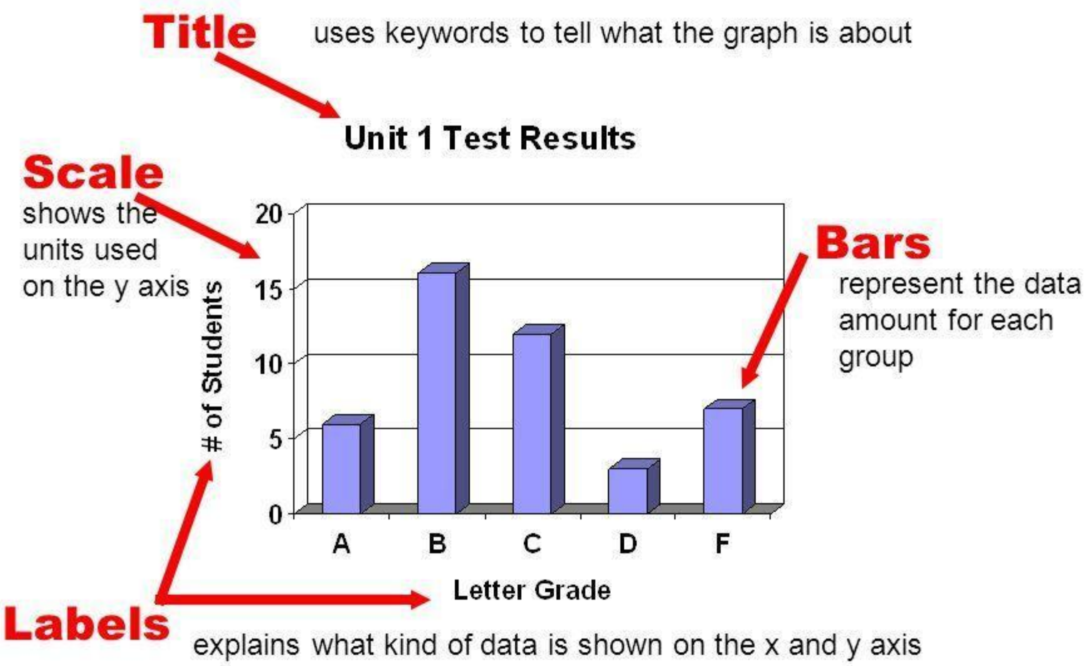


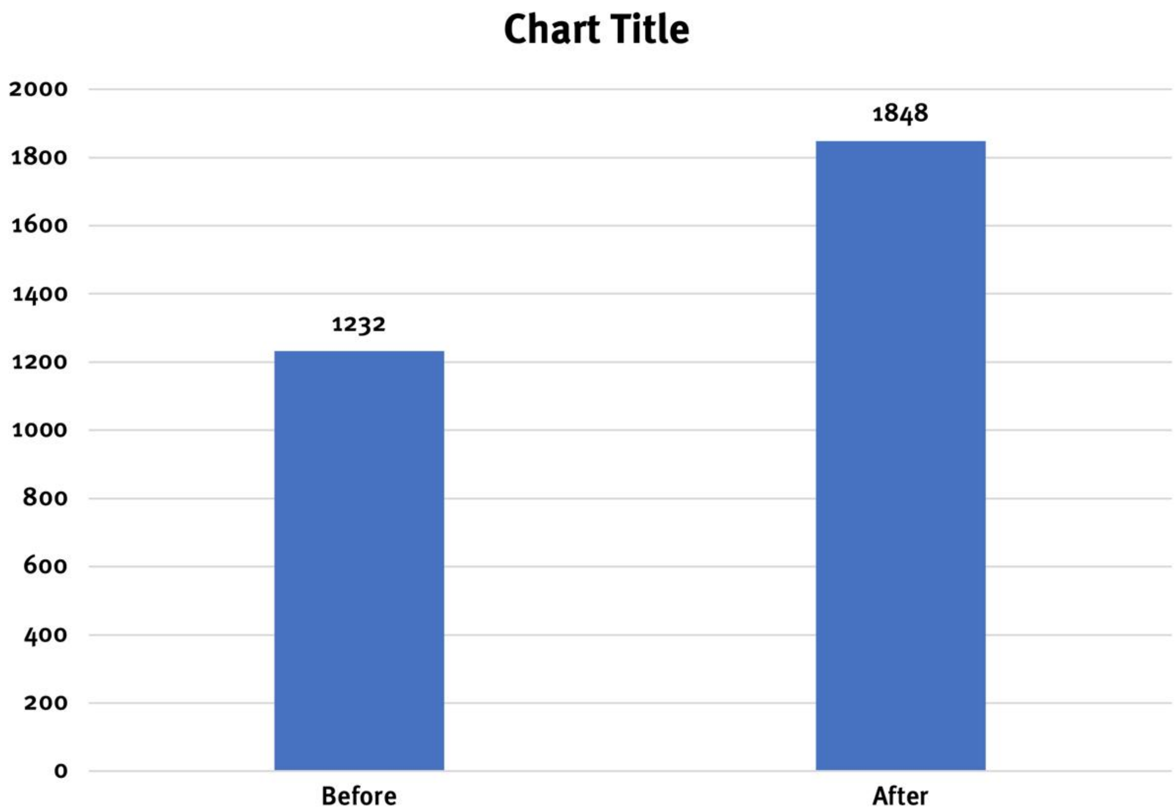
Grouped Bar Graph

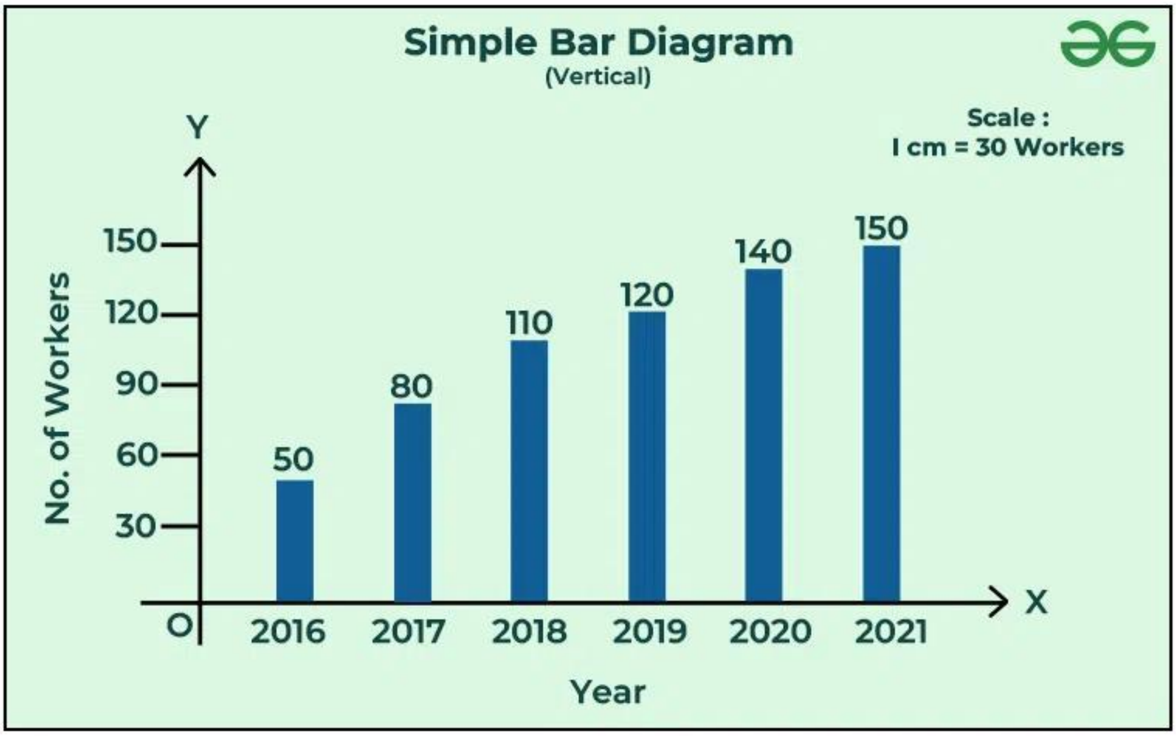

Vertical bar graph
70
60
50
40
30
20
10
0
price
one
two
three
four
five
pen sold


Types of Bar Charts
Basic Bar Chart
Grouped Bar Chart
Stacked Bar Chart
100% Stacked Bar Chart
Text A
Text B
Text C
Text D

Parts of a Bar Graph





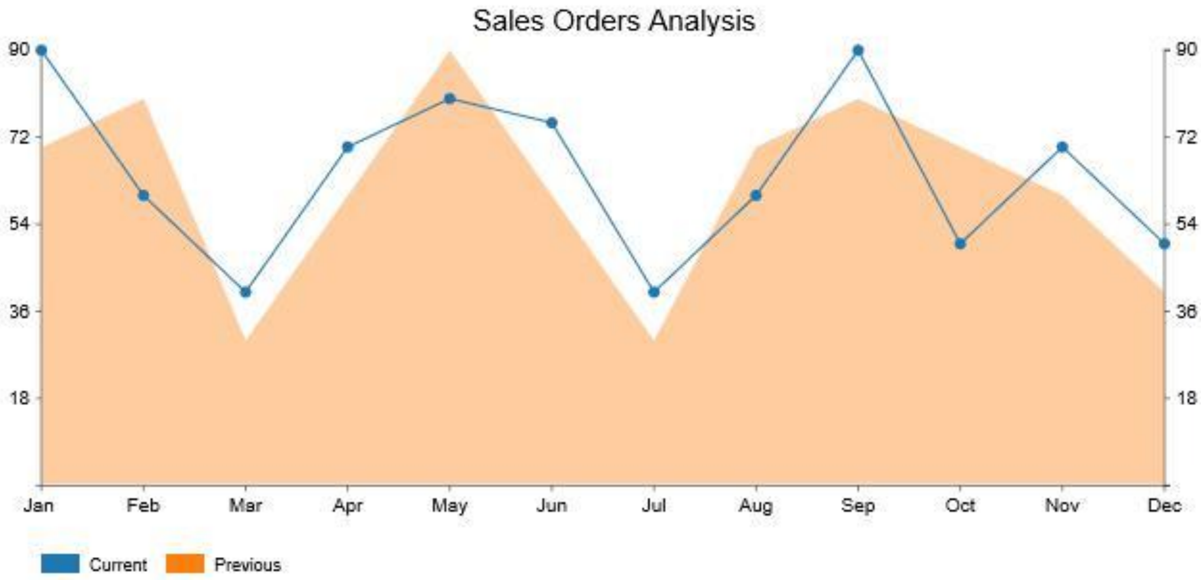


Use

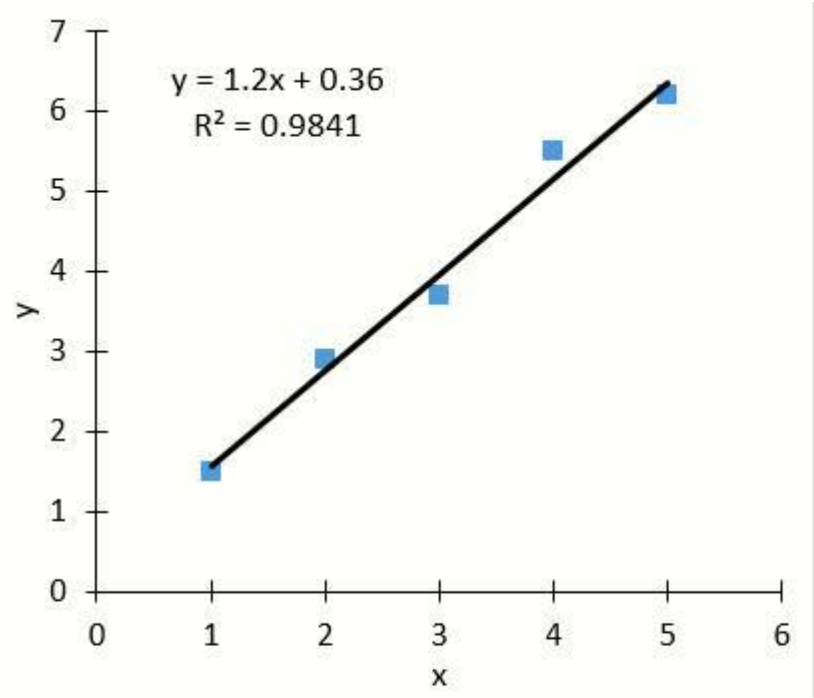
Compare categories

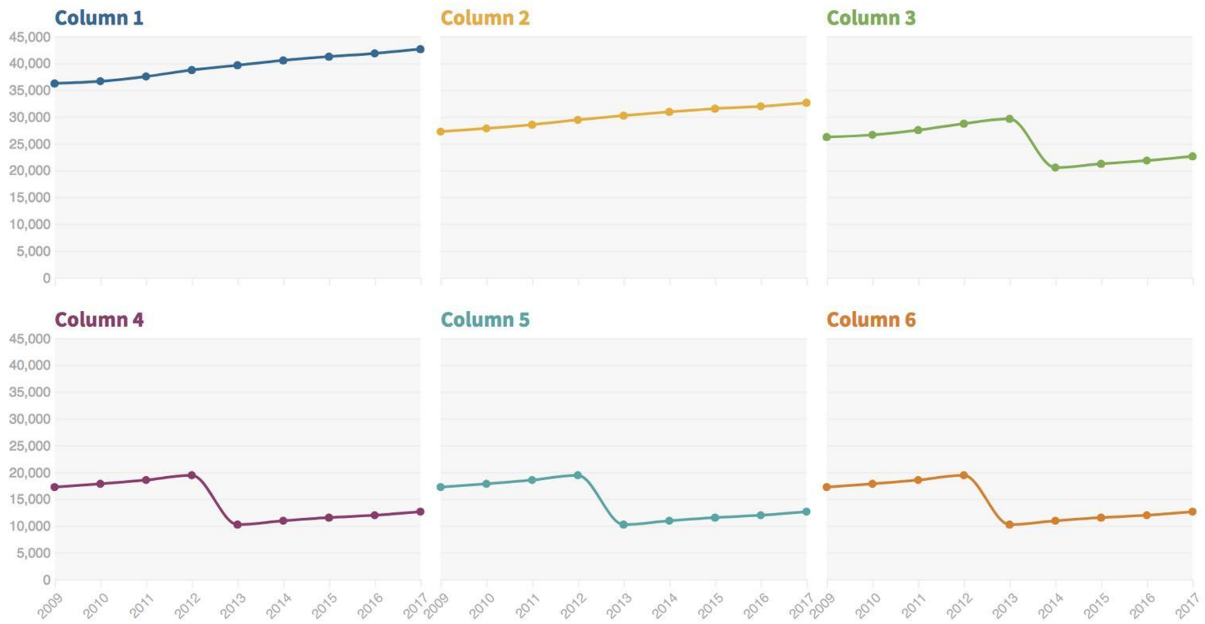
Line Chart

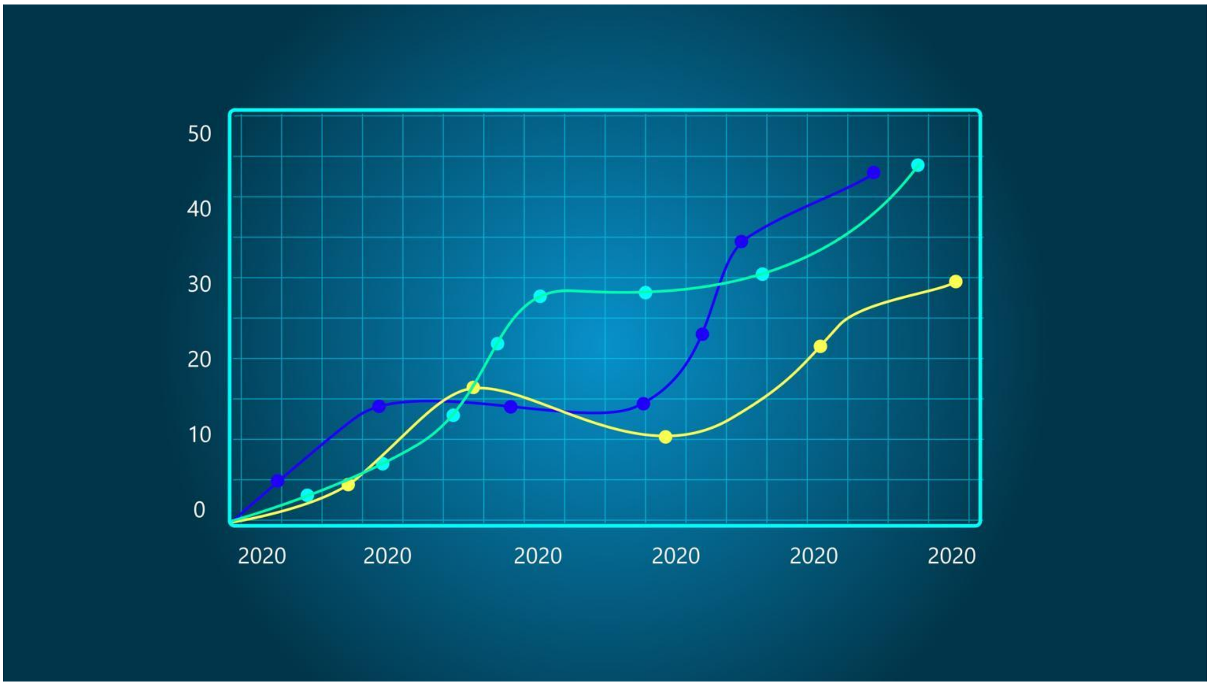
1
0
2011
2012
2013
2014
2015
2016
2017
2018

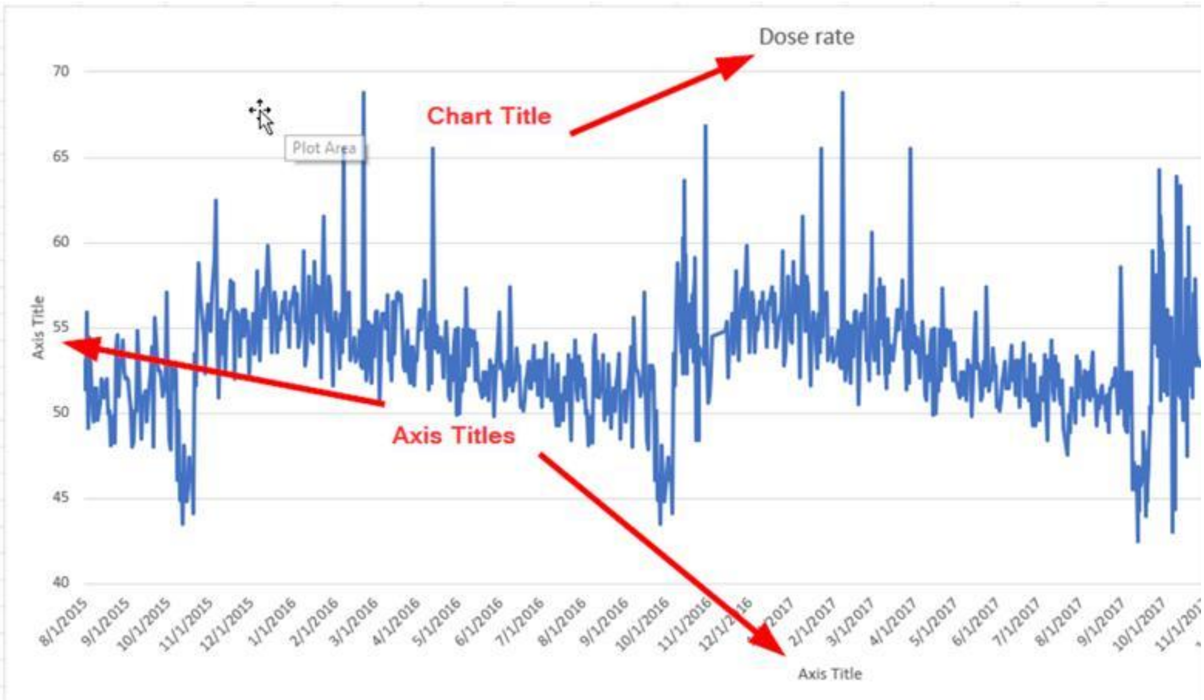


Sales Orders Analysis
90
72
54
36
18
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Current
Previous


y = 1.2x + 0.36
R² = 0.9841
y
x
0
1
2
3
4
5
6
7


Column 1
Column 2
Column 3
Column 4
Column 5
Column 6
45,000
40,000
35,000
30,000
25,000
20,000
15,000
10,000
5,000
0
2009
2010
2011
2012
2013
2014
2015
2016
2017


50
40
30
20
10
0
2020
2020
2020
2020
2020
2020


Dose rate
Chart Title
Plot Area
Axis Titles
Axis Title
Axis Title
70
65
60
55
50
45
40
8/1/2015
9/1/2015
10/1/2015
11/1/2015
12/1/2015
1/1/2016
2/1/2016
3/1/2016
4/1/2016
5/1/2016
6/1/2016
7/1/2016
8/1/2016
9/1/2016
10/1/2016
11/1/2016
2/1/2017
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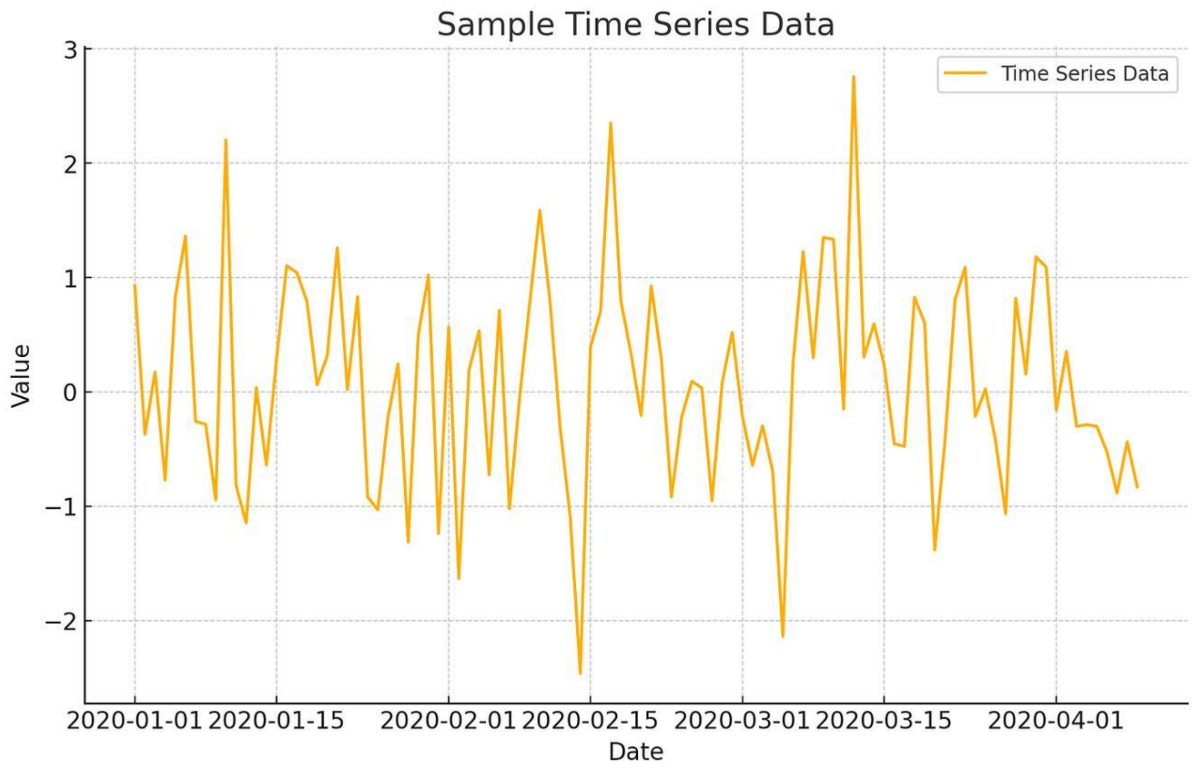
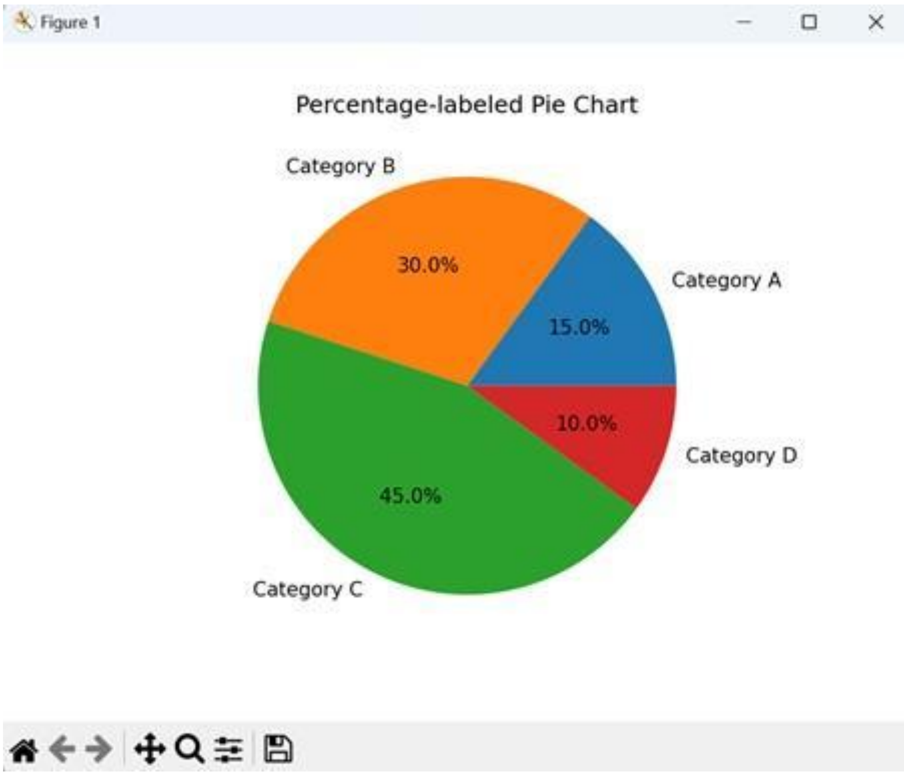
8/1/2017
9/1/2017
10/1/2017

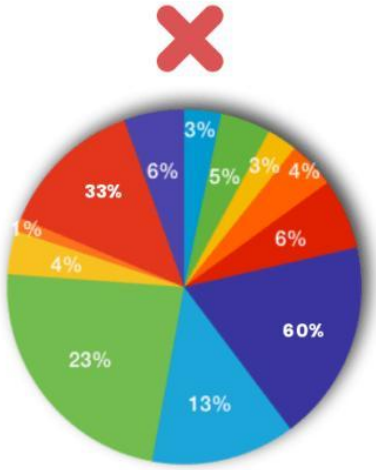


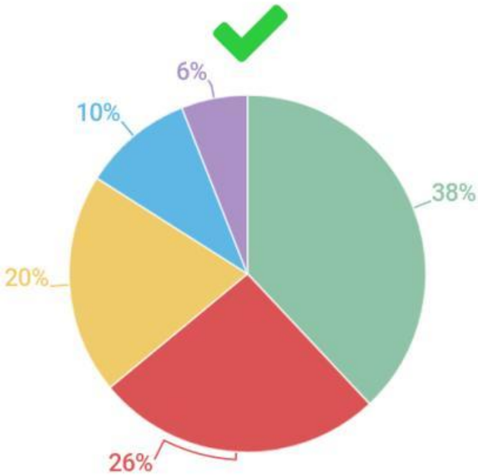
Use

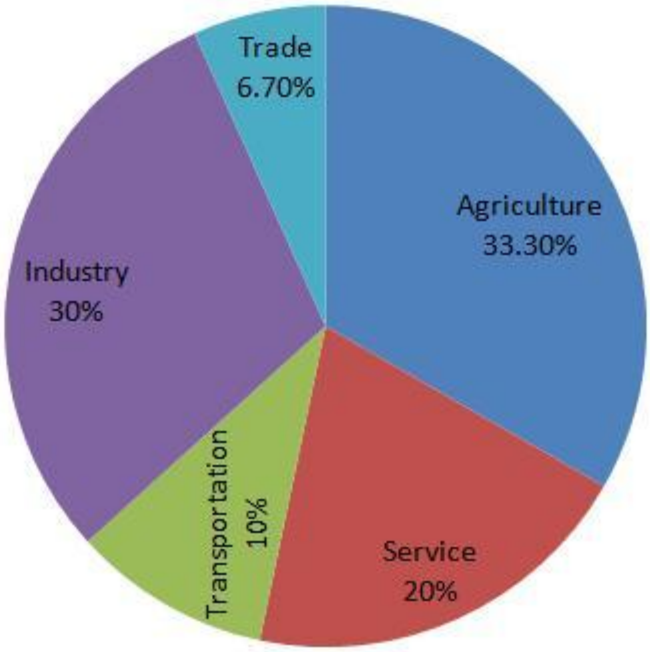
Show trends over time

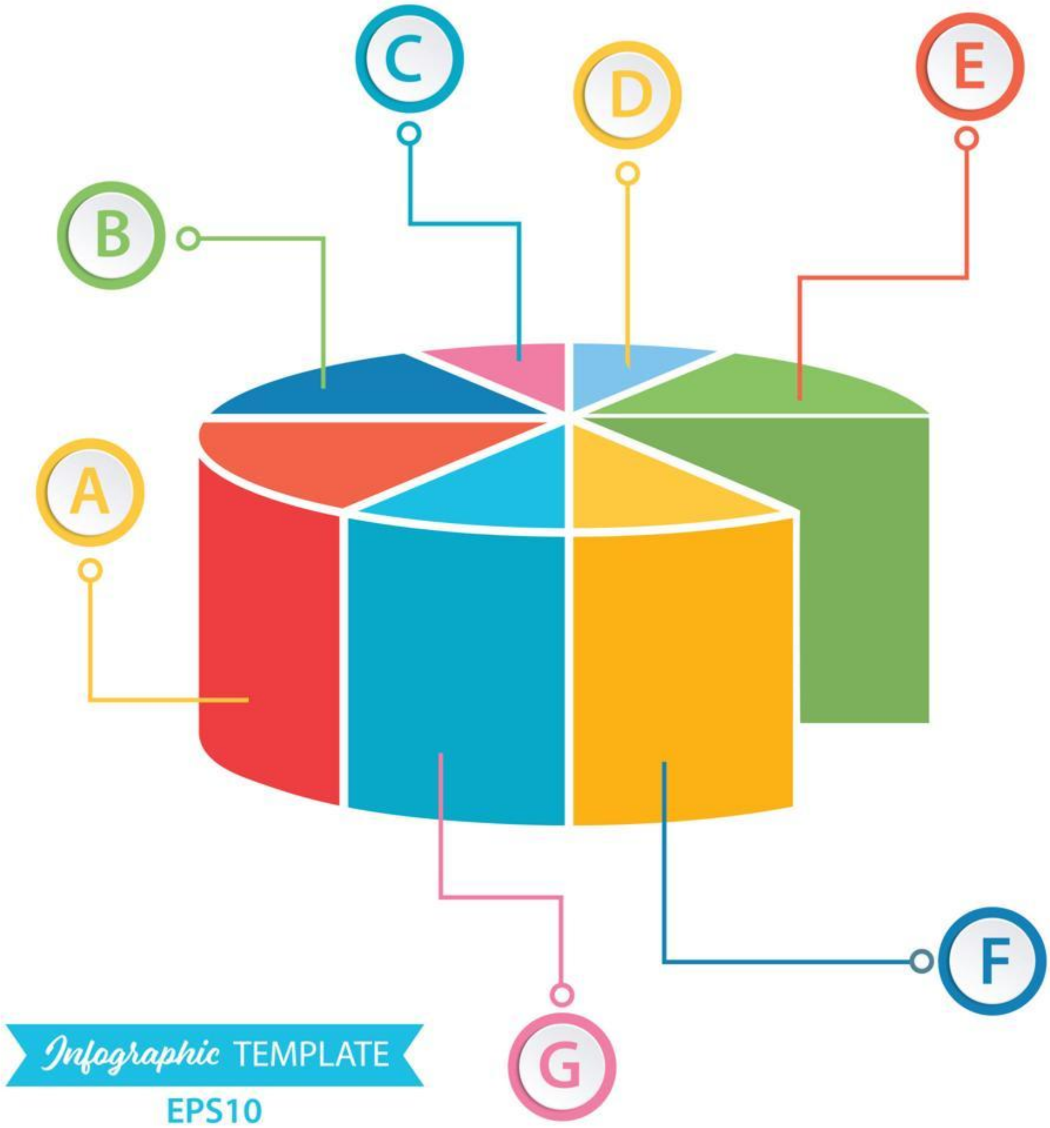
Pie Chart

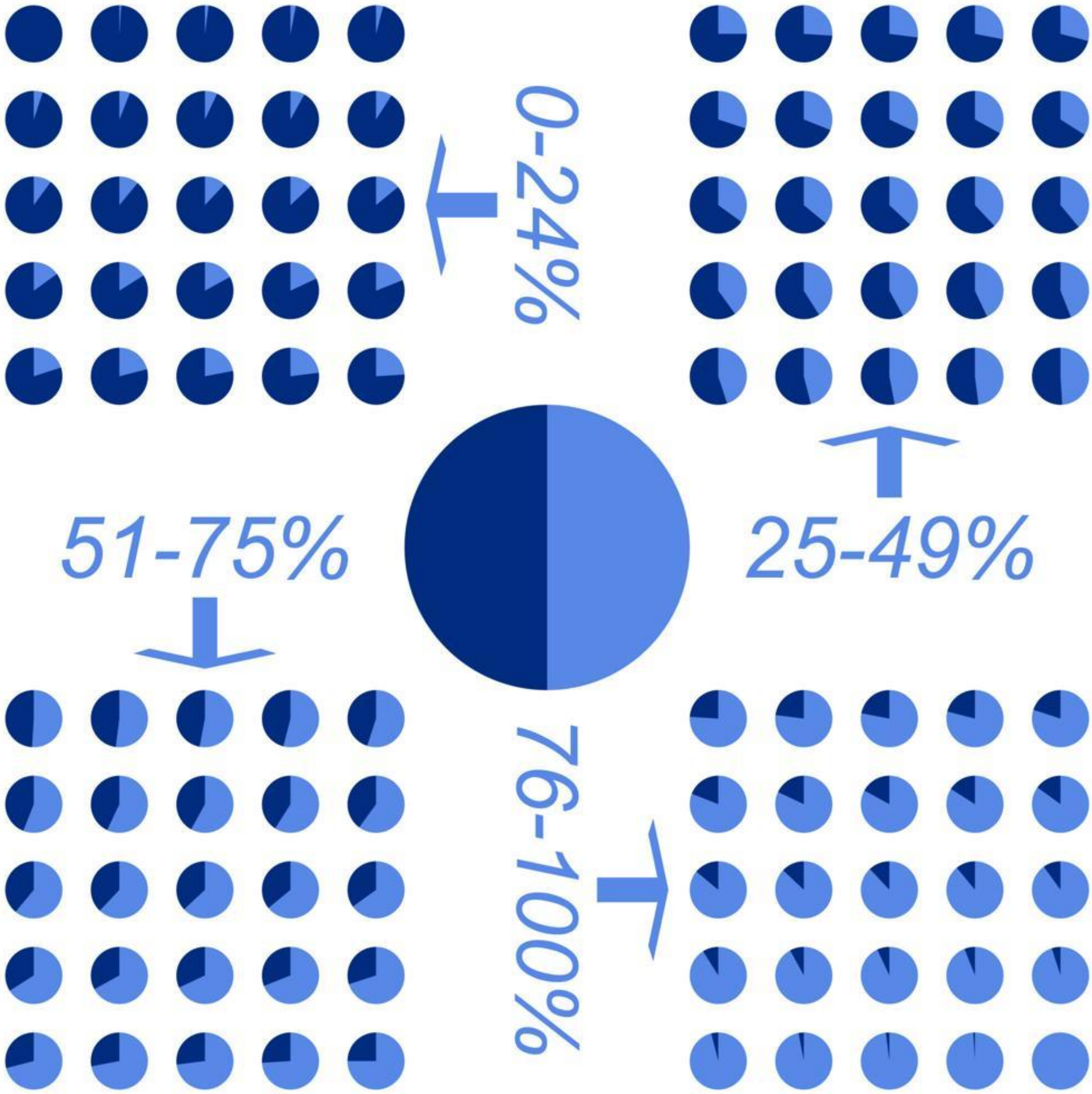


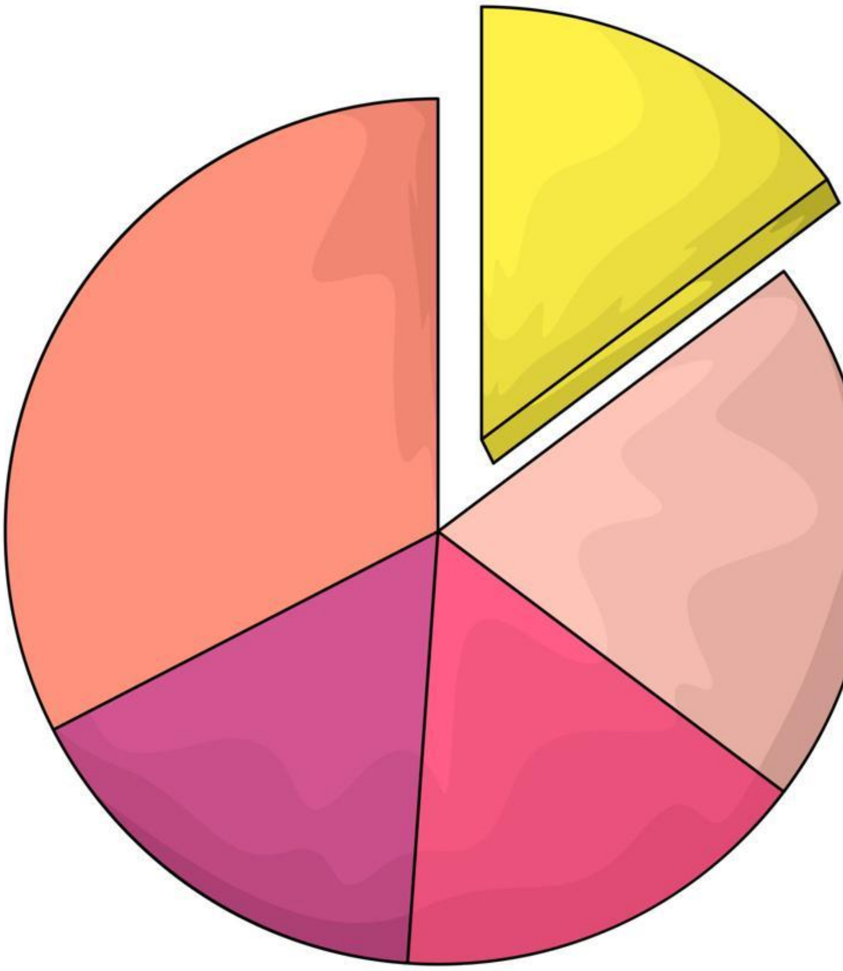

3%
6%
5%
3%
4%
33%
6%
4%
60%
23%
13%


6%
10%
38%
20%
26%


Trade
6.70%
Agriculture
33.30%
Industry
30%
Transportation
10%
Service
20%


A
B
C
D
E
F
G
Infographic TEMPLATE
EPS10


0-24%
25-49%
51-75%
76-100%



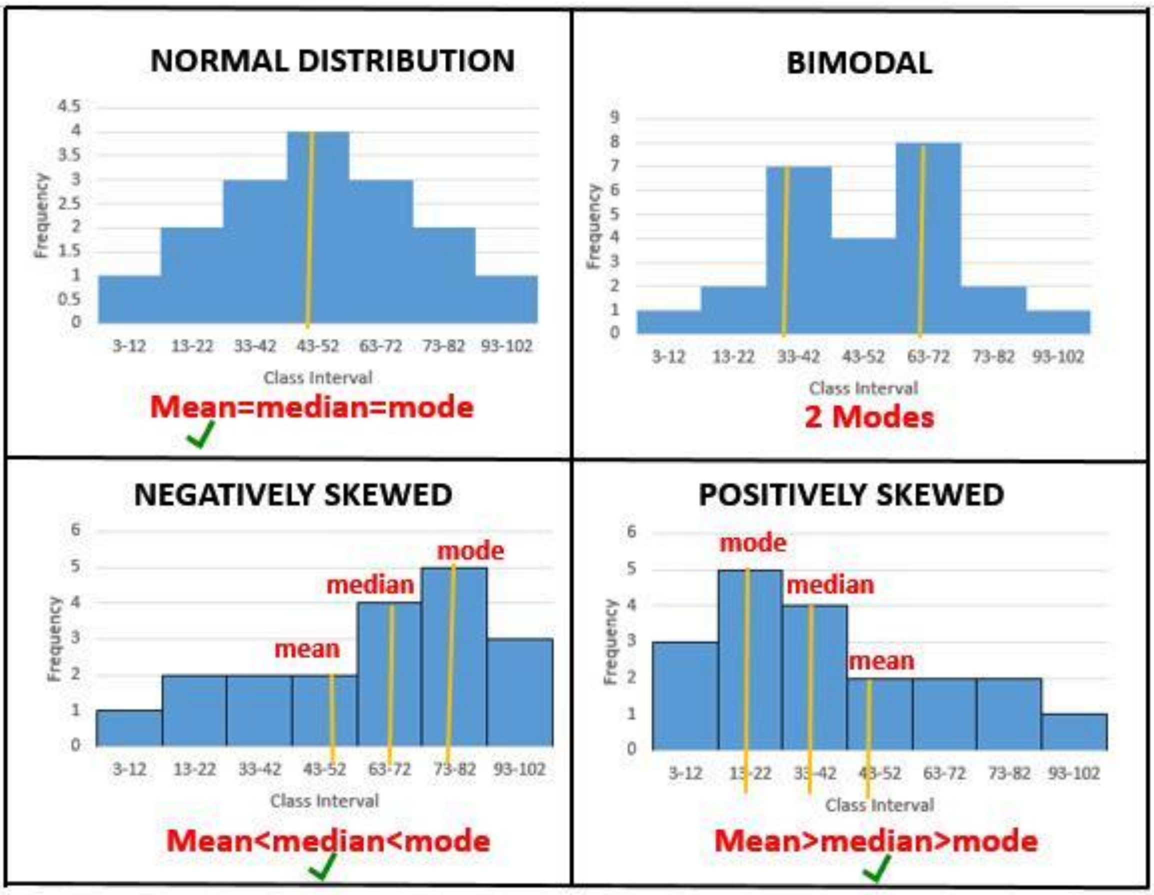
Use

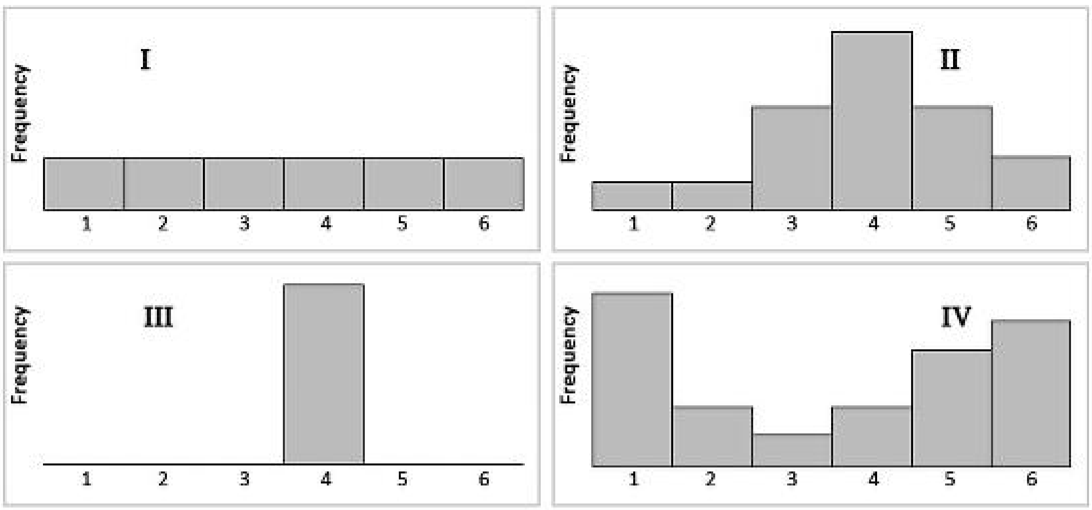
Show proportions

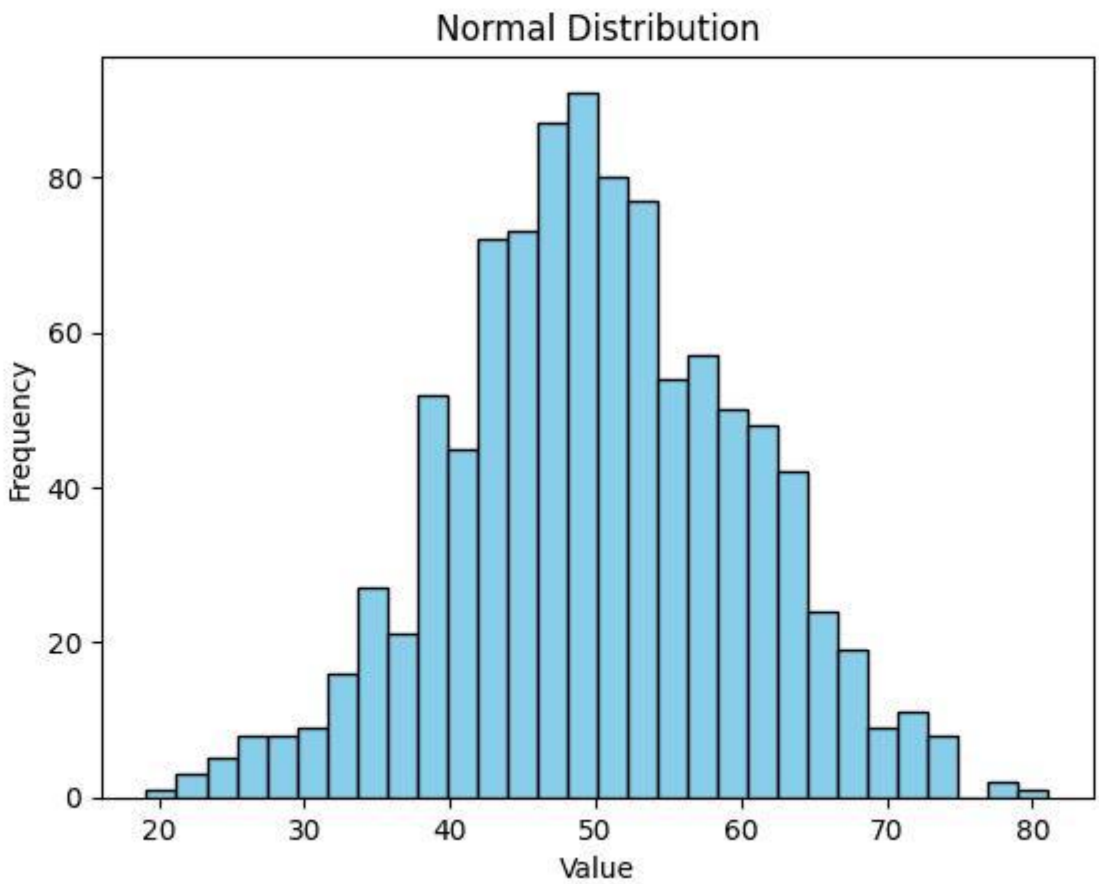
Reality

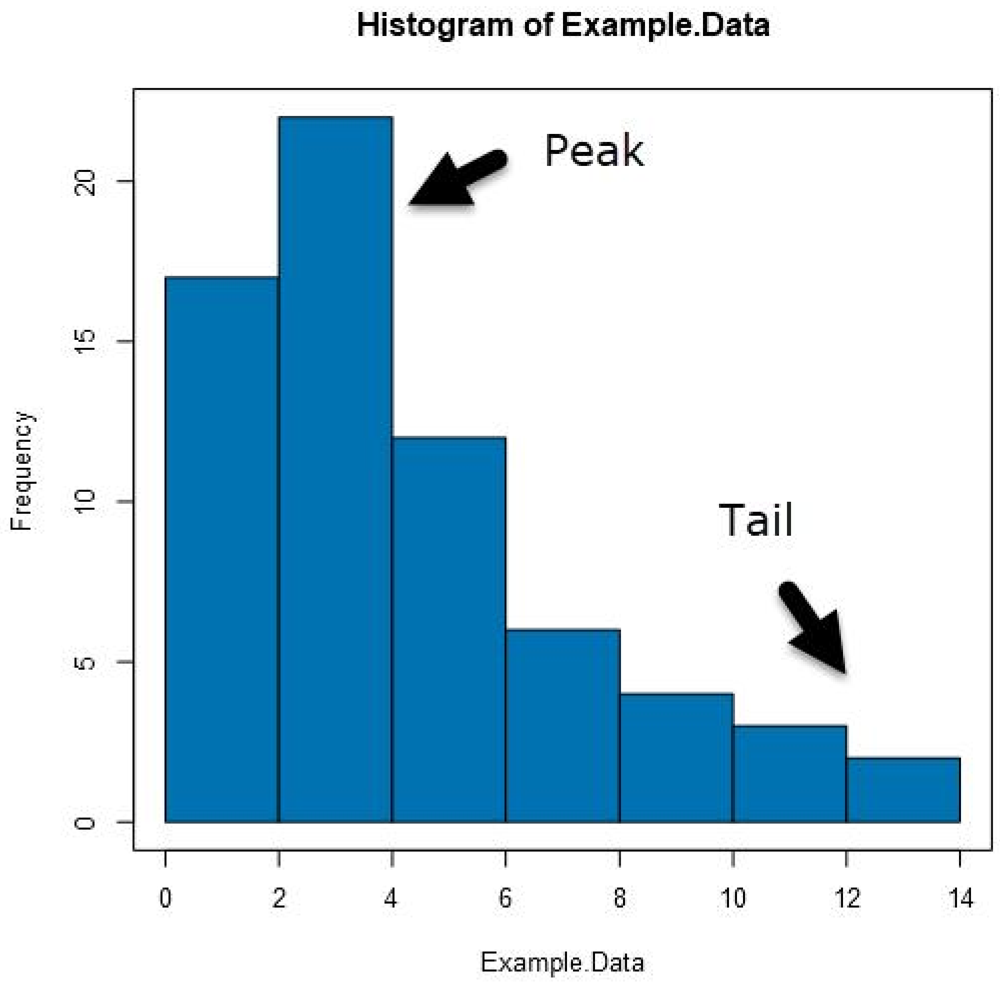
Pie charts are often misused. Humans compare angles poorly.

Histogram

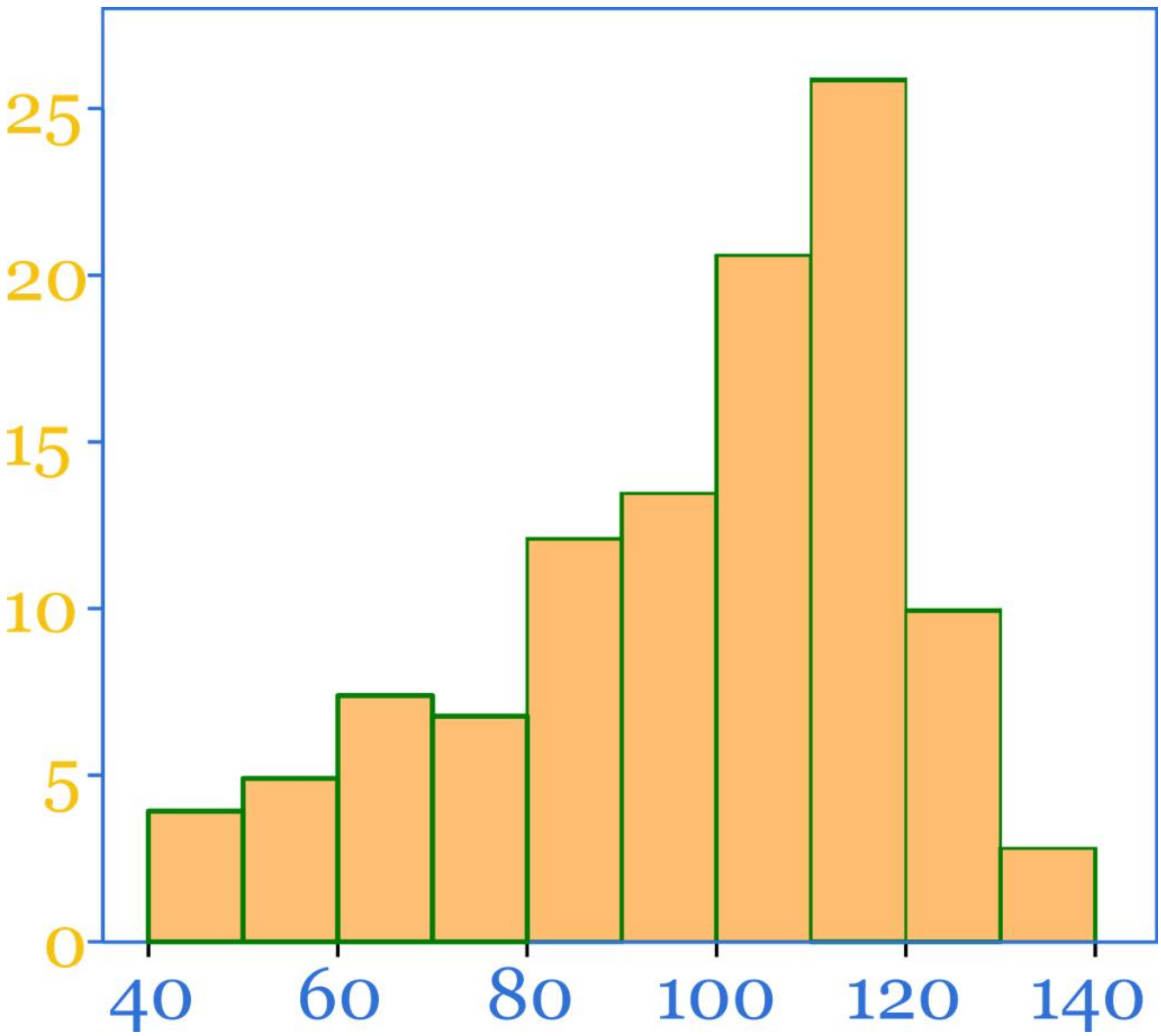

NORMAL DISTRIBUTION
Frequency
Class Interval
3-12
13-22
33-42
43-52
63-72
73-82
93-102
Mean=median=mode
BIMODAL
2 Modes
NEGATIVELY SKEWED
mean
median
mode
Mean<median<mode
POSITIVELY SKEWED
Mean>median>mode


I
II
III
IV
Frequency
1
2
3
4
5
6


Normal Distribution
Frequency
Value
0
20
40
60
80
20
30
40
50
60
70
80



Histogram of Example.Data
Peak
Tail
Frequency
0
5
10
15
20
0
2
4
6
8
10
12
14
Example.Data

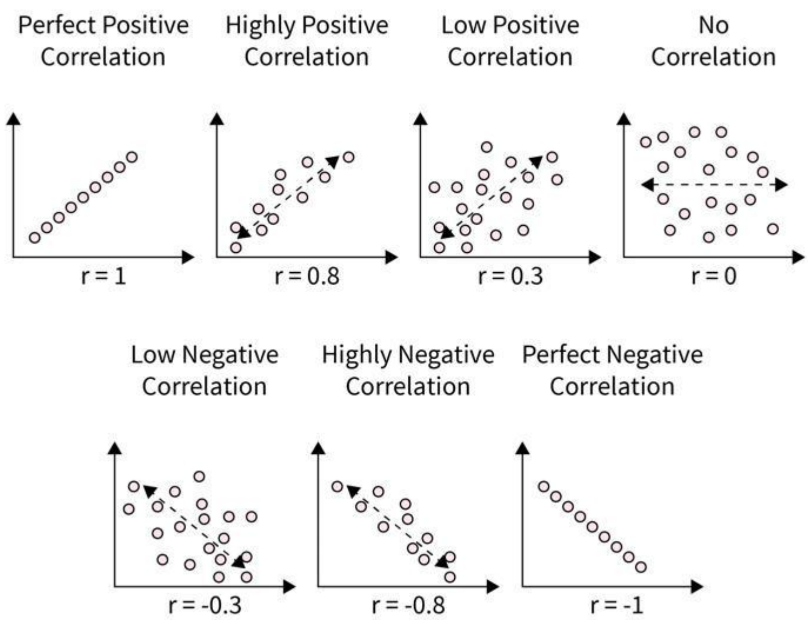


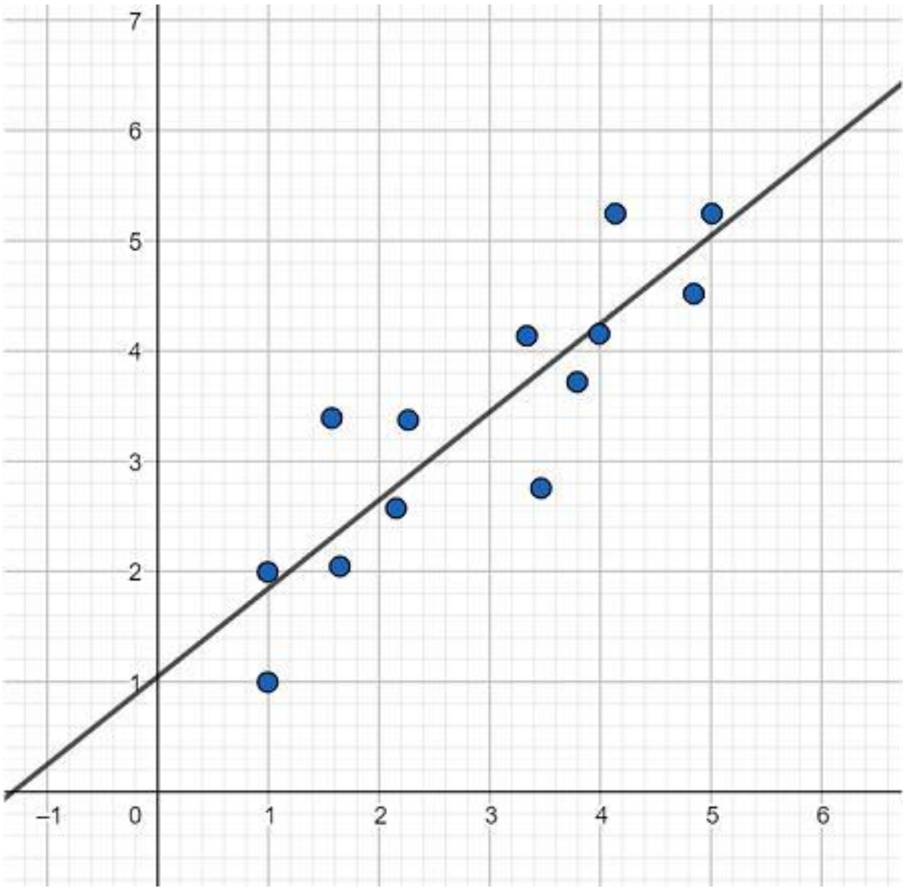
Use

Show distribution of data

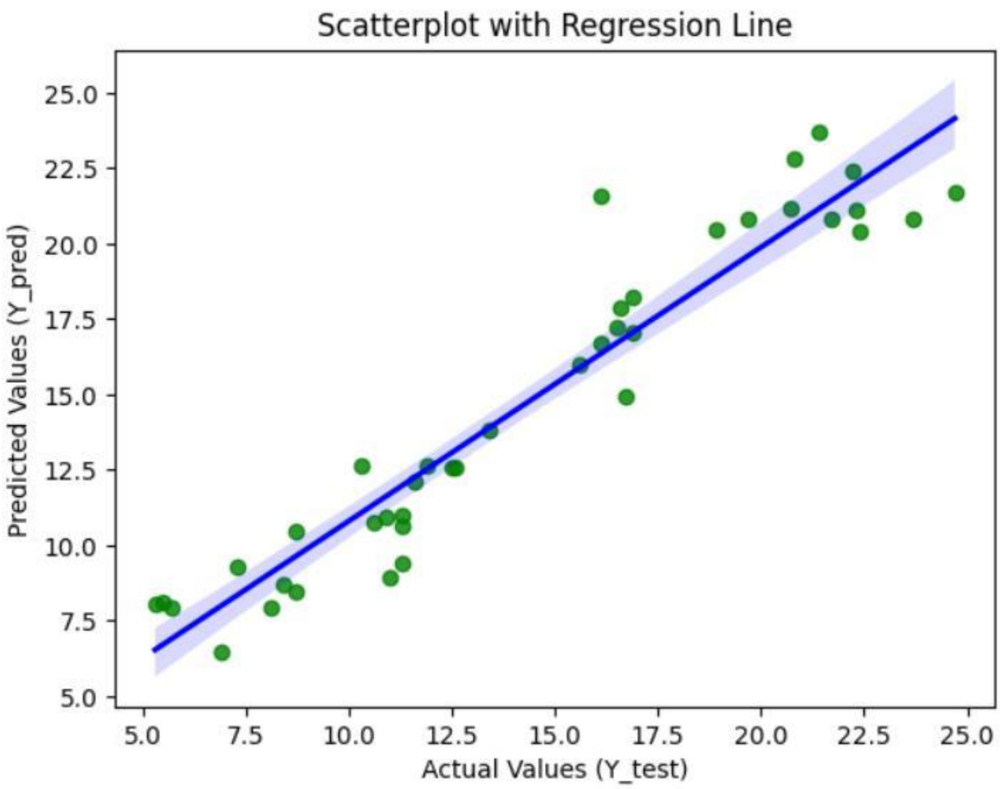
Scatter Plot

Scatter Plots & Correlation Examples

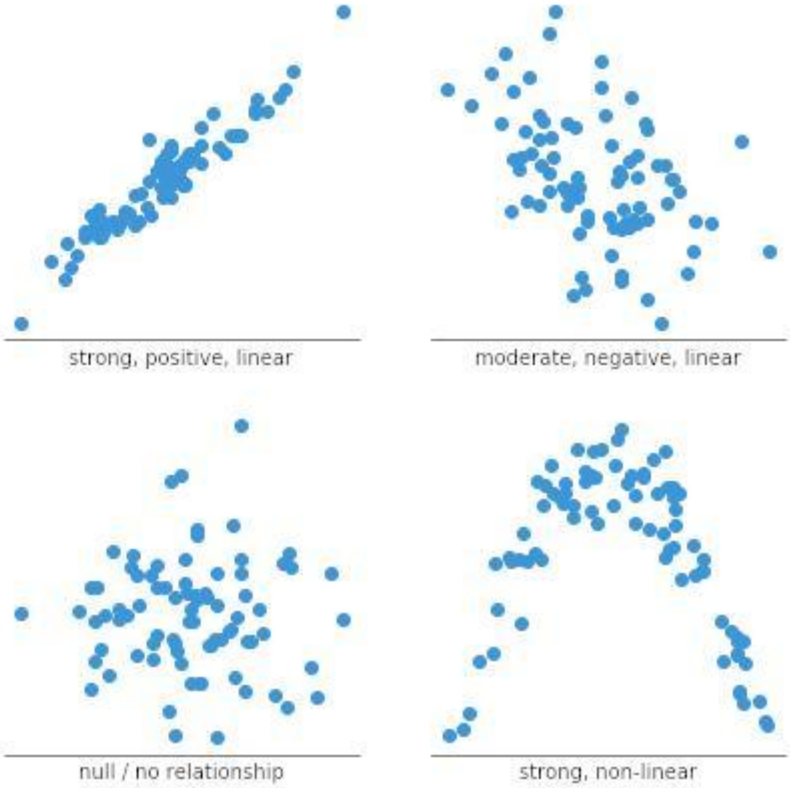


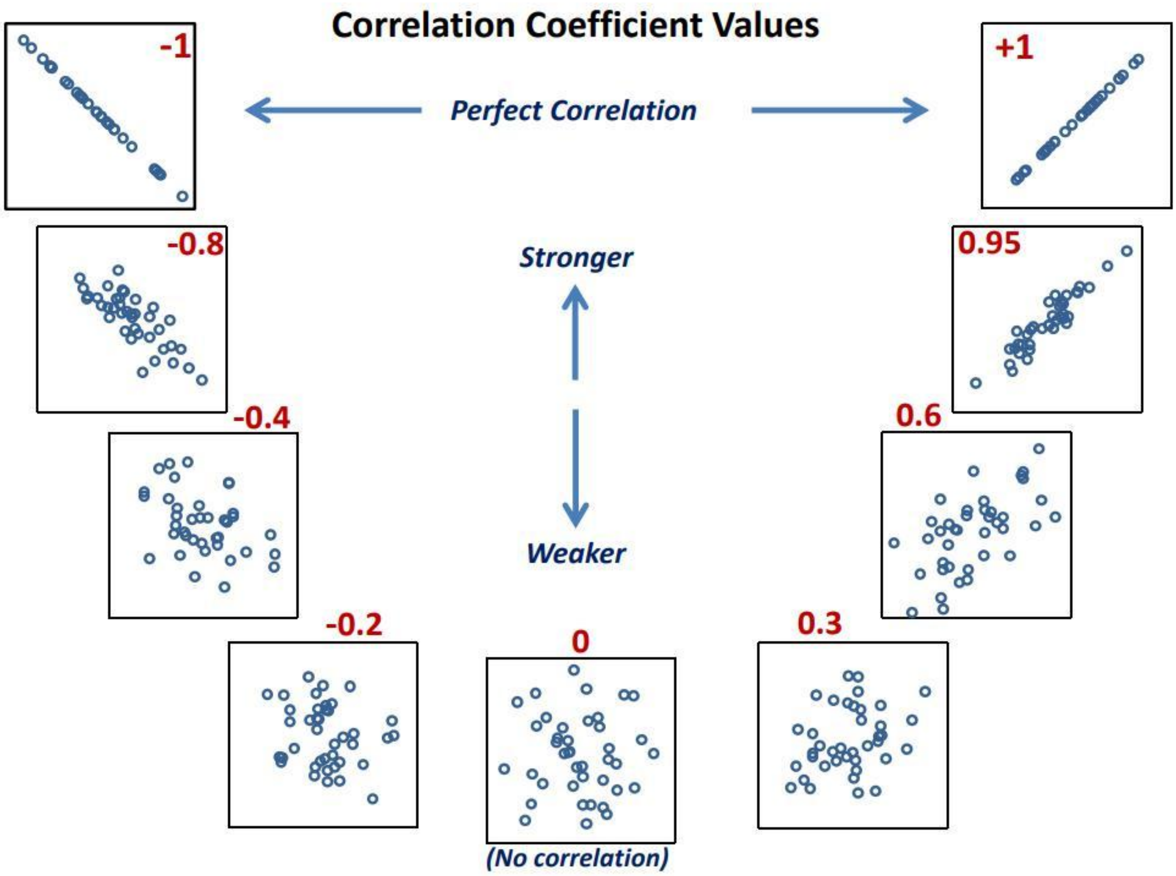


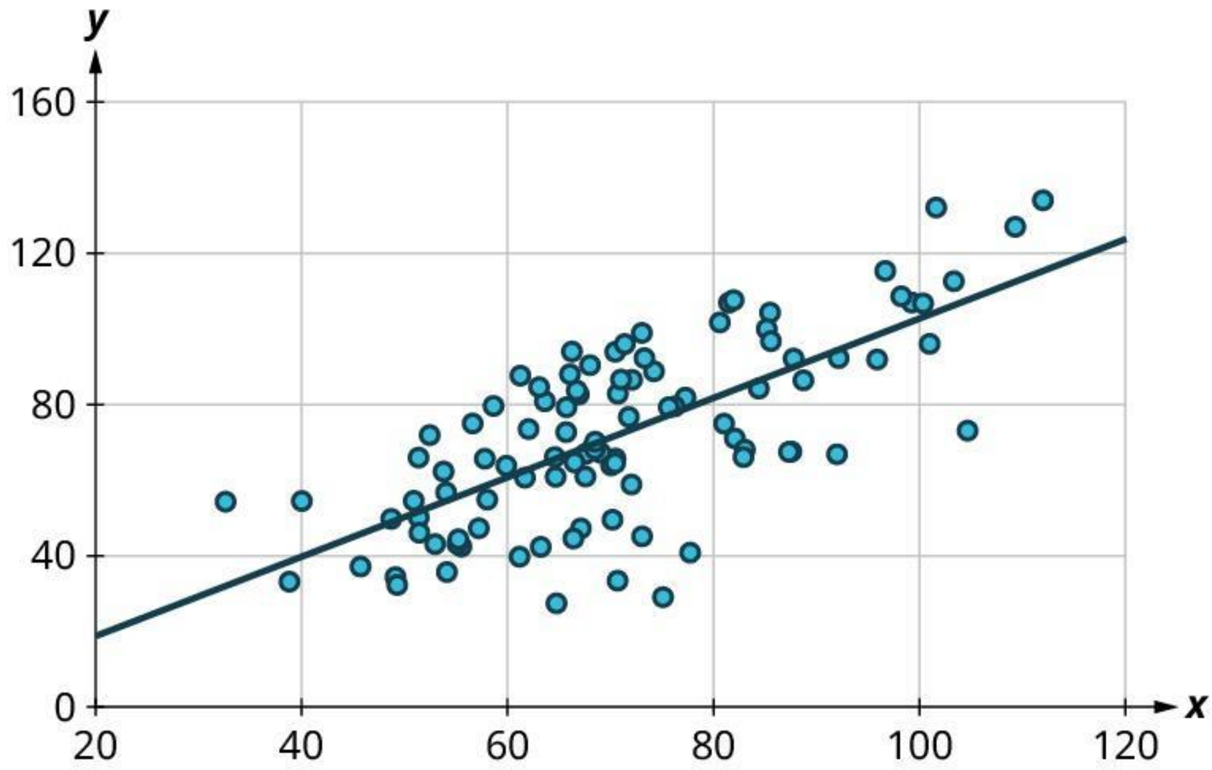
7
6
5
4
3
2
1
−1
0
1
2
3
4
5
6



Scatterplot with Regression Line
Predicted Values (Y_pred)
Actual Values (Y_test)
25.0
22.5
20.0
17.5
15.0
12.5
10.0
7.5
5.0
5.0
7.5
10.0
12.5
15.0
17.5
20.0
22.5
25.0


strong, positive, linear
moderate, negative, linear
null / no relationship
strong, non-linear


Correlation Coefficient Values
-1
Perfect Correlation
+1
-0.8
Stronger
0.95
-0.4
0.6
Weaker
-0.2
0
0.3
(No correlation)



Use

Show relationship between two variables

Critical insight

Wrong chart = wrong interpretation

5) Visualization Tools in Python

Matplotlib

Basic plotting library

Seaborn

Advanced statistical visualization

Example

```
import matplotlib.pyplot as plt

plt.plot([1,2,3], [10,20,30])
plt.show()
```

Seaborn Example

```
import seaborn as sns

sns.histplot(data=df, x="marks")
```

Difference

- Matplotlib → control
- Seaborn → simplicity + aesthetics

---

6) Common Visualization Mistakes

- Misleading scales
- Too many colors
- Wrong chart type
- Ignoring labels

PART 2: HADOOP

---

1) Introduction to Hadoop

What it actually is

Framework for storing and processing **big data** across multiple machines.

Why Hadoop exists

Single system cannot handle:

- Massive data
- High computation

Key idea

Distribute work across cluster

---

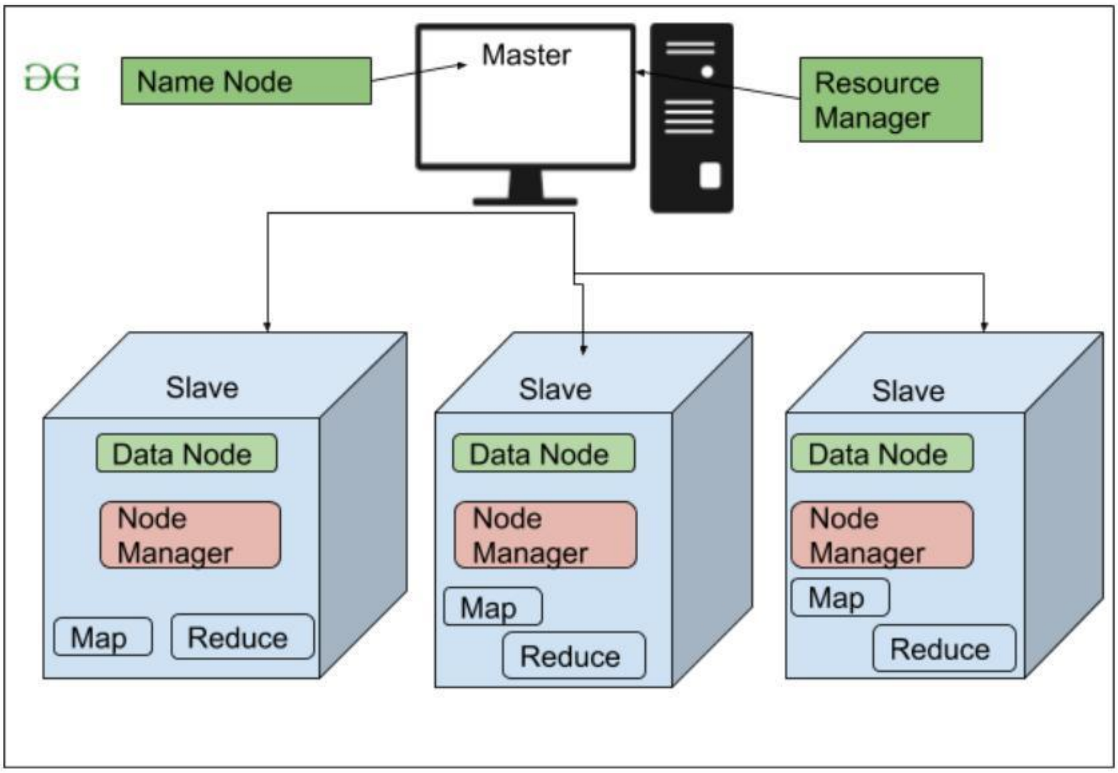
2) Hadoop Ecosystem

Core components:

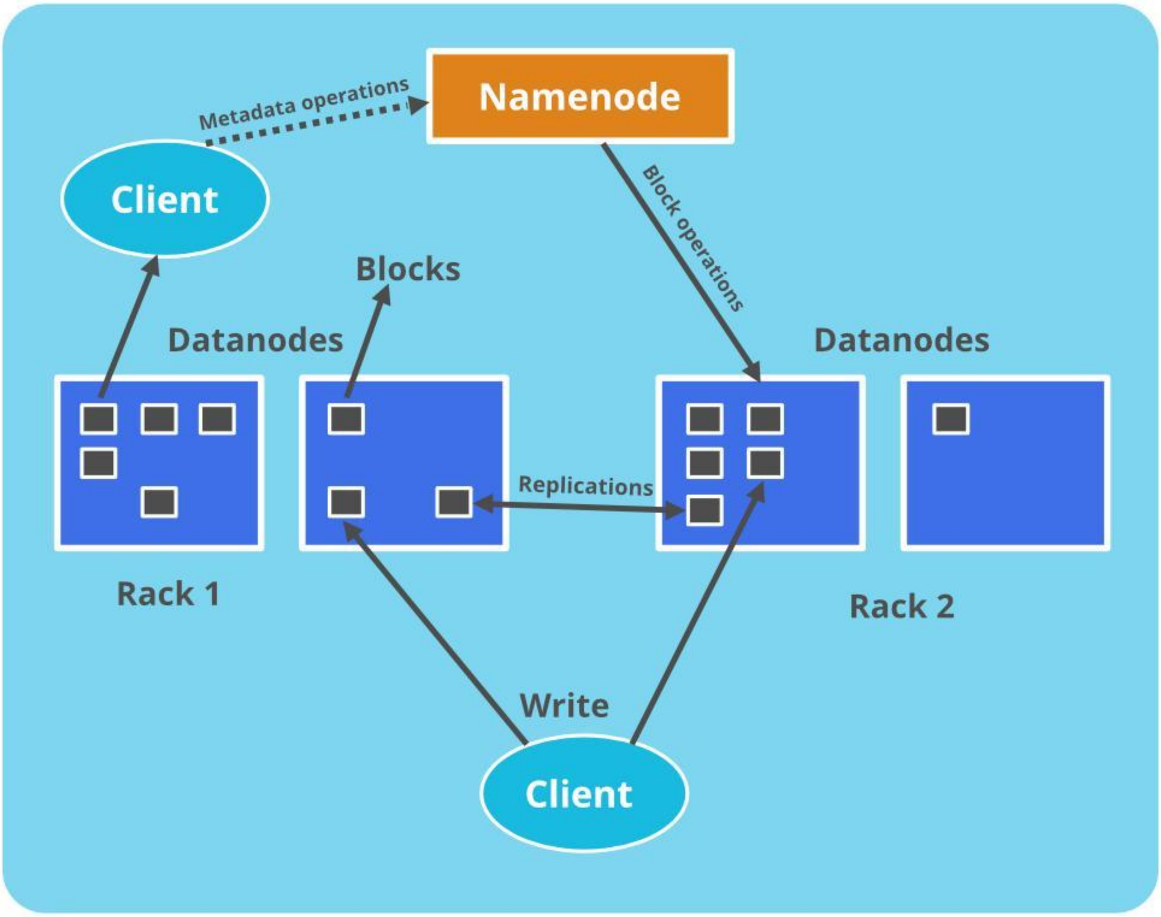
HDFS (Hadoop Distributed File System)

Stores data across multiple machines.

Architecture

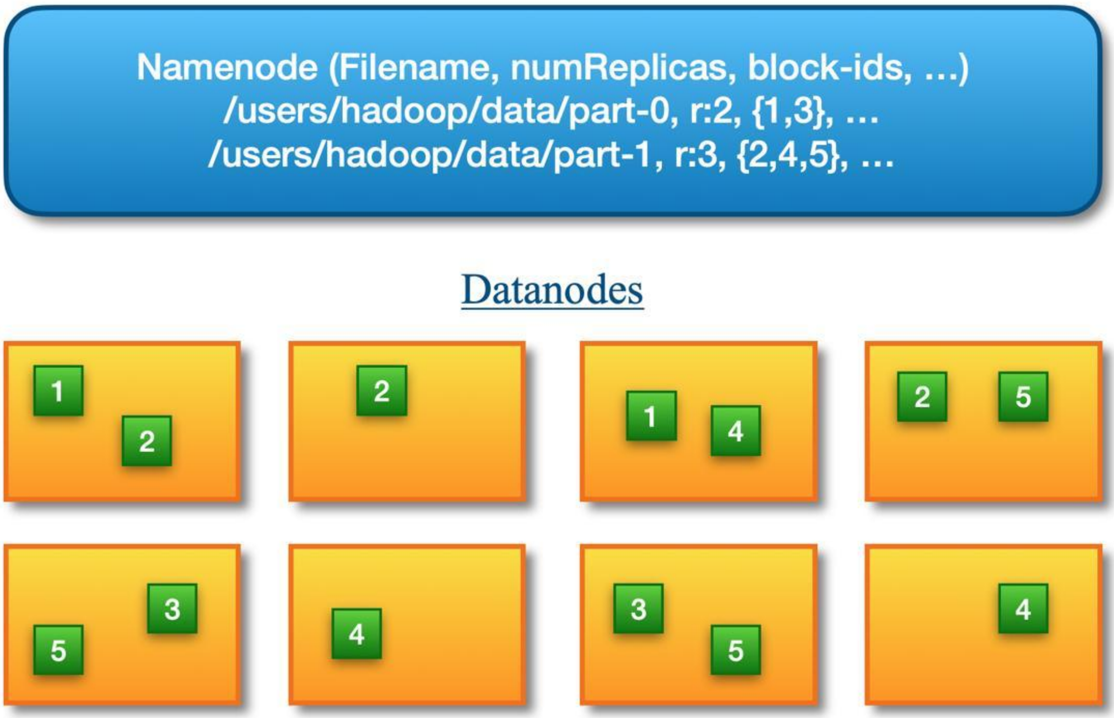


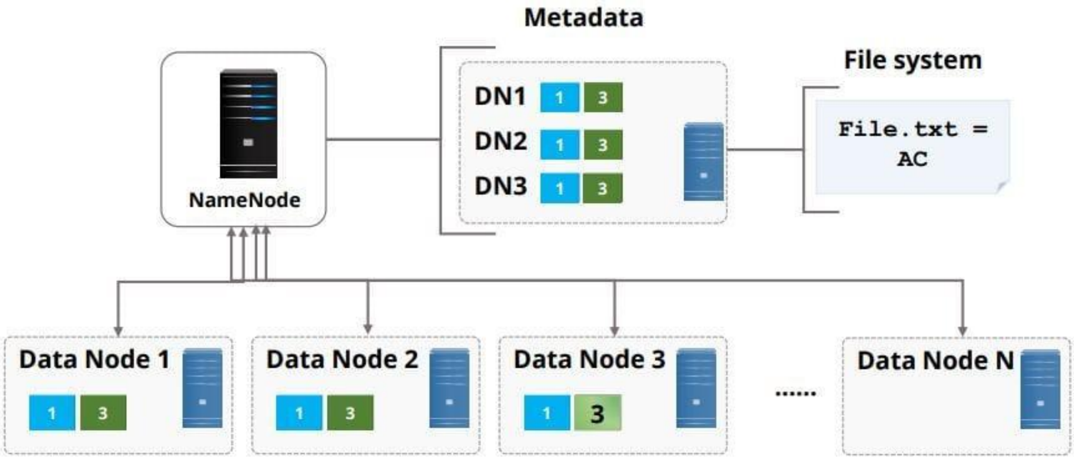
Name Node
Master
Resource Manager
Slave
Data Node
Node Manager
Map
Reduce
Slave
Data Node
Node Manager
Map
Reduce
Slave
Data Node
Node Manager
Map
Reduce

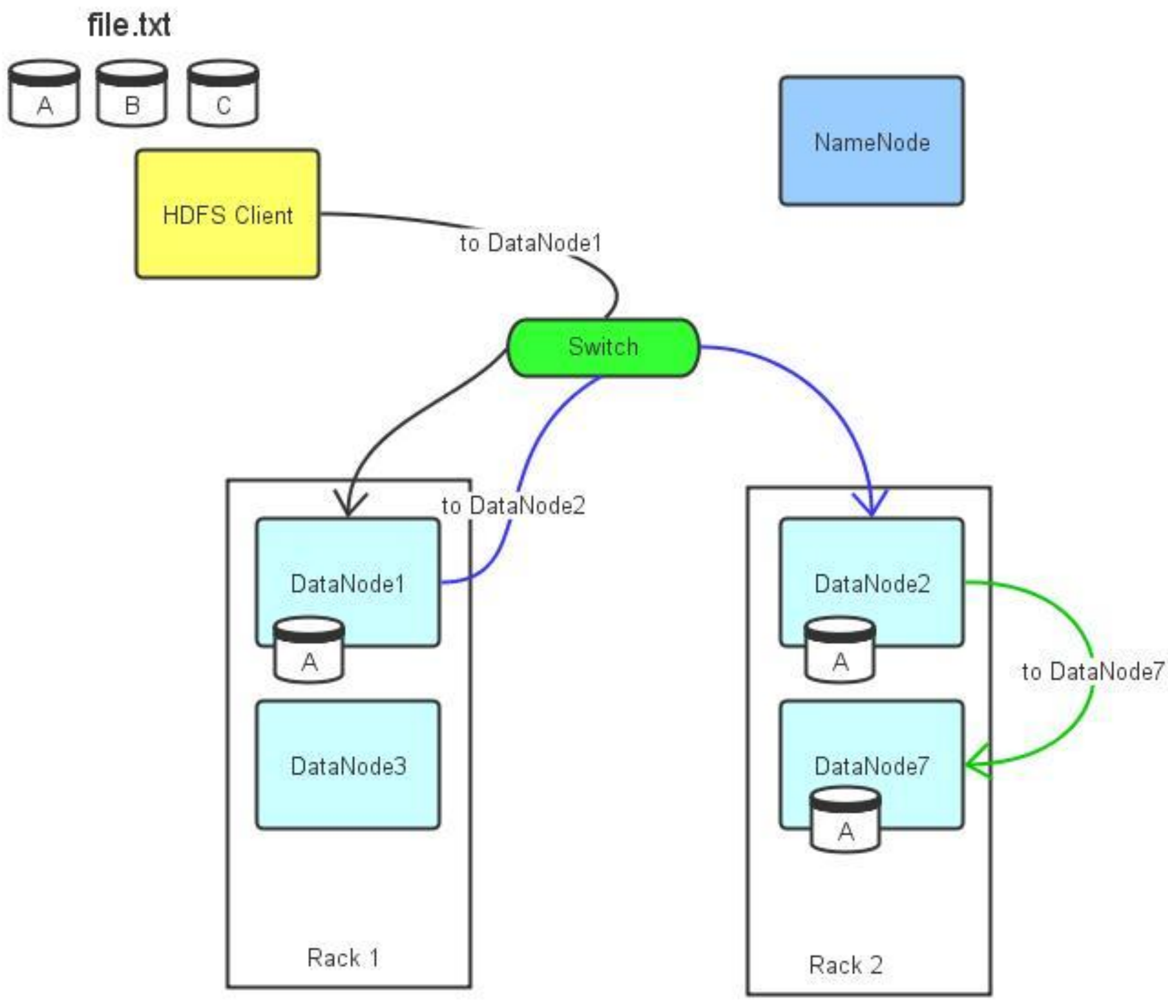


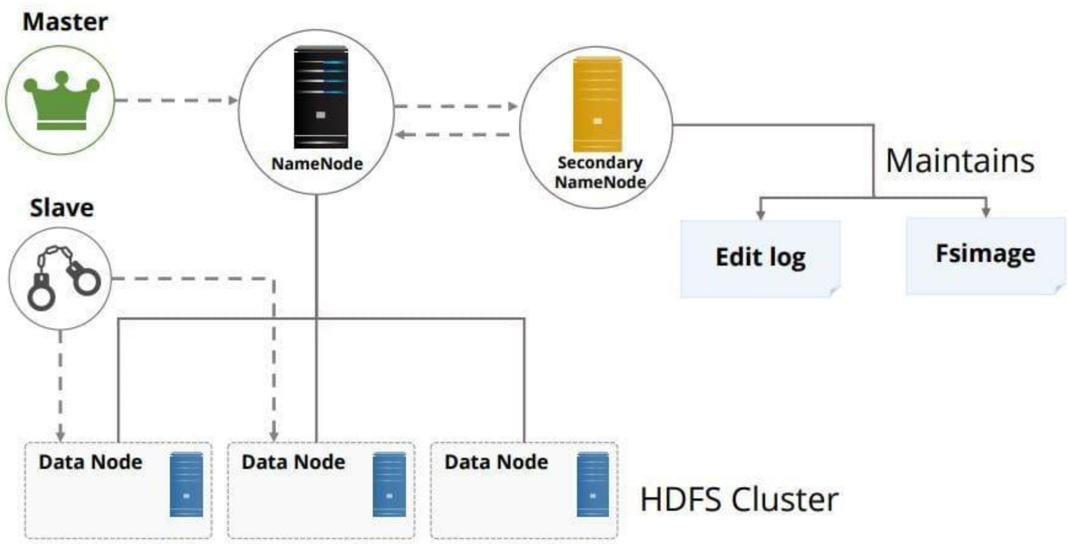
Namenode
Metadata operations
Client
Block operations
Blocks
Datanodes
Datanodes
Replications
Rack 1
Rack 2
Write
Client

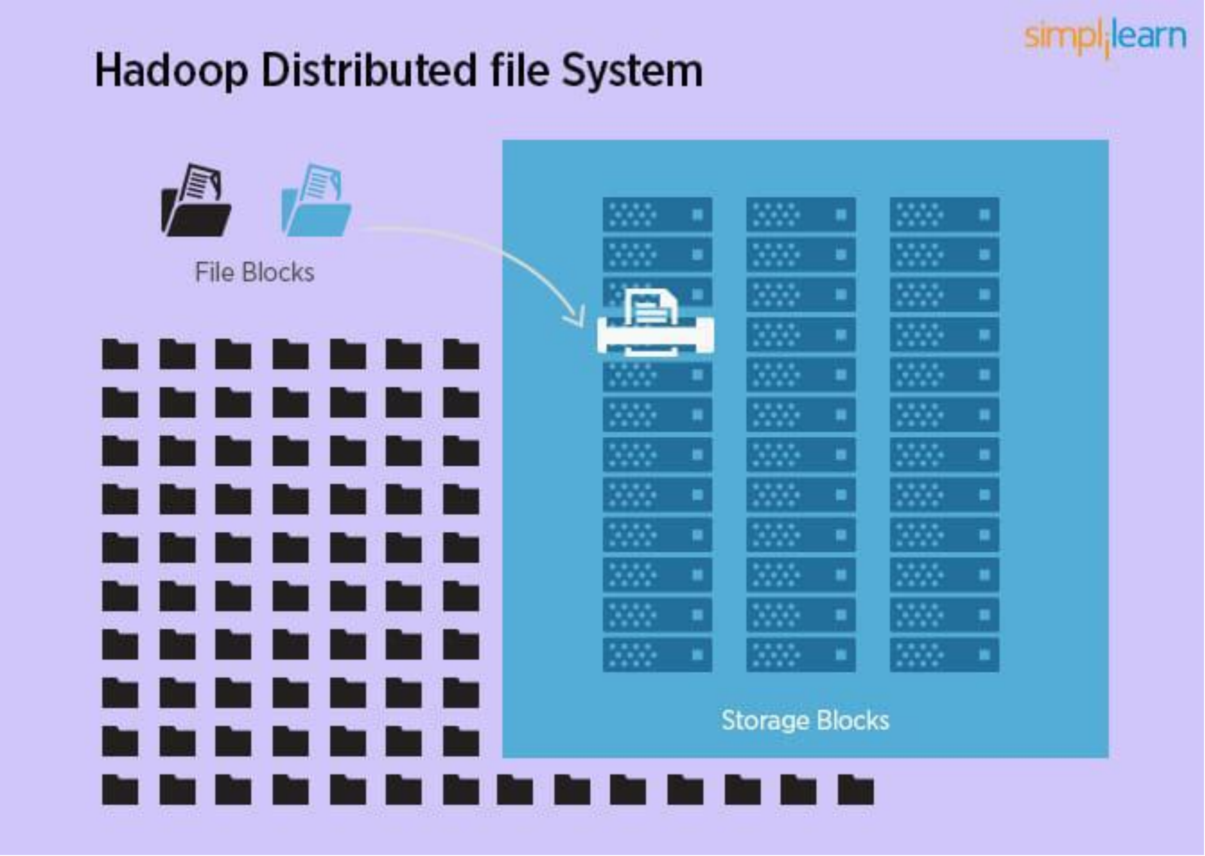
Block Replication






file.txt
A
B
C
HDFS Client
NameNode
to DataNode1
Switch
to DataNode2
DataNode1
A
DataNode3
Rack 1
DataNode2
A
to DataNode7
DataNode7
A
Rack 2


Master
NameNode
Secondary NameNode
Maintains
Slave
Edit log
Fsimage
Data Node
Data Node
Data Node
HDFS Cluster



Components

- NameNode → manages metadata
- DataNode → stores actual data

Key concept

Data is split into blocks and distributed.

Replication

Data is copied across nodes to avoid loss.

---

3) MapReduce

Processing model in Hadoop.

Idea

Break big task into smaller tasks.

Steps

1. Map → process data
2. Shuffle → group data

3. Reduce → aggregate results

Example

Word count:

- Map → (word, 1)
- Reduce → sum counts

Reality

MapReduce is slow compared to modern tools but conceptually important.

---

4) YARN (Yet Another Resource Negotiator)

Manages resources in Hadoop.

Role

- Allocates CPU and memory
- Schedules jobs

---

5) Hadoop vs Traditional Systems

Hadoop	Traditional
Distributed	Centralized
Scalable	Limited
Fault tolerant	Failure prone

---

6) Advantages of Hadoop

- Handles huge data
- Fault tolerant
- Cost-effective

---

7) Limitations

- Slow for real-time processing
- Complex setup
- Not suitable for small data

---

8) Example from Syllabus

Task:

- Visualize dataset

- Understand HDFS architecture

Visualization Example

import seaborn as sns

sns.scatterplot(x="sepal_length", y="petal_length", data=df)

Hadoop Understanding

Explain:

- Data storage → HDFS
- Processing → MapReduce

Critical Fail Points

If you say:

- “Pie chart is best for all cases”
- “Hadoop is just storage”

You don’t understand this unit.

PRACTICAL 1 — DATA WRANGLING (IRIS / CSV DATASET)

(From your sheet)

Q1. What is Data Wrangling and why is it required?

Answer:
Data wrangling is the process of cleaning, transforming, and organizing raw data into a usable format.

Raw data is:

- incomplete
- inconsistent
- noisy

Models cannot work on raw data. If you skip wrangling, your model learns wrong patterns.

Q2. Why do we use pandas instead of basic Python lists?

Answer:
Pandas provides:

- tabular structure (DataFrame)
- fast vectorized operations
- built-in functions for cleaning and analysis

Lists cannot handle structured data efficiently and require manual loops, which are slower.

Q3. What is a DataFrame?

Answer:
A DataFrame is a 2D labeled data structure with rows and columns.

- Columns → features
- Rows → records

It is similar to a database table but stored in memory.

Q4. Difference between Series and DataFrame?

Answer:

- Series → 1D (single column)
- DataFrame → 2D (multiple columns)

A DataFrame is a collection of Series.

Q5. What is the purpose of df.head() and df.tail()?

Answer:

- head() shows first 5 rows
- tail() shows last 5 rows

Used to quickly inspect dataset structure without printing full data.

Q6. What does df.shape represent?

Answer:
It returns:

(rows, columns)

Example:
(150, 5) → 150 records, 5 features

Q7. What is meant by data types in a dataset?

Answer:
Each column has a type:

- int → integer
- float → decimal
- object → string

Correct datatype is necessary because:

- models require numeric data
- operations depend on type

Q8. What are missing values and why are they problematic?

Answer:
Missing values are empty or null entries.

Problems:

- break calculations
- reduce model accuracy
- cause bias

Q9. Methods to handle missing values?

Answer:

1. Remove rows
2. Replace with mean
3. Replace with median
4. Replace with mode

Choice depends on data distribution.

Q10. What are outliers?

Answer:
Outliers are extreme values far from other data points.

Example:
10, 12, 11, 500 → 500 is outlier

Q11. Why are outliers dangerous?

Answer:
They:

- distort mean
- mislead models
- create false patterns

Q12. What is feature scaling?

Answer:
Scaling adjusts values to a similar range.

Example:

- Age: 20–60
- Salary: 10,000–1,00,000

Without scaling, large values dominate model learning.

Q13. What is normalization?

Answer:
Transforms data into range [0,1].

Used when:

- features have different scales
- distance-based algorithms are used

Q14. What is discretization?

Answer:
Converting continuous values into categories.

Example:
Age → Young, Adult, Old

Q15. Why is data wrangling the most time-consuming step?

Answer:
Because:

- real-world data is messy
- requires manual decisions
- involves multiple iterations

Most effort is spent fixing data, not building models.

Reality Check

If you answer like:

- “Wrangling means cleaning data”
- “Outliers are bad values”

You will get cross-questioned and fail.

You must explain:

- **why** each step is done
- **when** to apply each technique

PRACTICAL 2 — DATA WRANGLING (STUDENT PERFORMANCE DATASET)

(From your sheet)

Q1. What kinds of inconsistencies can exist in a student dataset?

Answer:
Inconsistencies are logical or formatting errors in data.

Examples:

- Different formats: "Male" vs "M"
- Invalid values: marks = 150 out of 100
- Mixed types: numeric column stored as string
- Duplicate student entries

These create incorrect analysis because the model treats them as valid inputs.

Q2. How do you detect missing values in a dataset?

Answer:
Using pandas:

```
df.isnull().sum()
```

This returns count of missing values per column.

You can also use:

```
df.info()
```

to see non-null counts.

Detection is critical because missing values are silent errors.

Q3. When should you remove rows instead of filling missing values?

Answer:
Remove rows when:

- Missing data is very small (<5%)
- Missing values are random
- Filling would distort meaning

Do NOT remove rows when:

- dataset is small
- missing values are significant

Blind deletion leads to data loss and bias.

Q4. Why is median preferred over mean in some cases?

Answer:
Median is robust to outliers.

Example:
Marks: 40, 42, 45, 100
Mean = 56.75 (misleading)
Median = 43.5 (accurate)

So, median is used when data is skewed.

Q5. How do you identify outliers in student data?

Answer:

Methods:

1. **Z-score method**

- Values far from mean (±3 std dev)

2. **IQR method**

- Q1, Q3
- Outliers = values outside:

Q1 - 1.5*IQR, Q3 + 1.5*IQR

3. **Visualization**

- Boxplot

Q6. Should outliers always be removed?

Answer:
No. That is a common mistake.

Remove only if:

- it is an error
- it is irrelevant

Keep if:

- it is real data
- it represents important behavior

Example:
Top student scoring 100 is not an outlier to remove.

Q7. What is feature scaling and why is it needed here?

Answer:
Feature scaling brings all variables to a comparable range.

Example:

- Study hours: 1–10
- Marks: 0–100

Without scaling:

- model gives more weight to marks

Q8. Difference between normalization and standardization?

Answer:

Normalization

Range: [0,1]
Used when distribution is unknown

Standardization

Mean = 0, Std = 1
Used when data is normally distributed

Q9. Why convert non-linear relationships into linear ones?

Answer:
Some models (like linear regression) assume linear relationships.

If data is non-linear:

- model underfits
- predictions are inaccurate

Transformation (log, square, etc.) makes pattern learnable.

Q10. What is encoding and why is it needed?

Answer:
Encoding converts categorical values into numeric form.

Example:

- Gender: Male → 0, Female → 1

Models cannot process text directly.

Q11. What problems arise from improper encoding?

Answer:
If encoding is wrong:

- model assumes false order

Example:

Low=1, Medium=2, High=3

This implies ranking even if categories are unrelated.

Q12. What is data transformation in this context?

Answer:
Changing format or scale of data.

Examples:

- Scaling
- Log transformation
- Encoding

Goal:
Make data suitable for model learning.

Q13. What happens if you skip data preprocessing?

Answer:

- Model learns incorrect patterns
- Accuracy becomes unreliable
- Predictions fail on new data

In short: model becomes useless.

Q14. How do you justify your preprocessing choices?

Answer:
Each step must be based on data characteristics:

- Missing values → choose mean/median based on distribution
- Outliers → check if error or valid
- Scaling → required for distance-based models

You must explain **why**, not just apply methods.

Q15. Why is documenting preprocessing steps important?

Answer:
Because:

- ensures reproducibility
- helps debugging
- allows model reuse

Without documentation, results cannot be trusted or repeated.

PRACTICAL 3 — DESCRIPTIVE STATISTICS & VISUALIZATION

(From your sheet)

Q1. What is descriptive statistics?

Answer:
Descriptive statistics summarizes and describes the main features of a dataset.

It does not make predictions. It only answers:

- What is the average?
- How spread is the data?
- What pattern exists?

Q2. Why do we calculate mean, median, and mode together?

Answer:
Because each captures a different aspect of central tendency.

- Mean → overall average
- Median → resistant to outliers
- Mode → most frequent value

Comparing them reveals distribution shape.

Q3. What does it mean if mean > median?

Answer:
Data is **right-skewed (positively skewed)**.

Reason:
Large values pull mean upward.

Example:
Income data where few people earn extremely high salaries.

Q4. What does it mean if mean < median?

Answer:
Data is **left-skewed (negatively skewed)**.

Reason:
Low values pull mean downward.

Q5. Why is standard deviation more useful than range?

Answer:
Range only considers two values:

- min and max

Standard deviation considers **all data points**, making it more reliable.

Range is easily distorted by a single outlier.

Q6. What does a high standard deviation indicate?

Answer:
Data points are widely spread from the mean.

Implication:

- high variability
- less consistency

Example:
Marks varying from 10 to 90

Q7. What does a low standard deviation indicate?

Answer:
Data points are close to the mean.

Implication:

- consistent data
- less variation

Q8. Why is variance squared?

Answer:
If we directly sum deviations:

- positive and negative values cancel out

So we square them to:

- make all values positive
- emphasize larger deviations

Q9. Why do we take square root in standard deviation?

Answer:
Variance is in squared units (e.g., $marks^2$), which is hard to interpret.

Taking square root:

- brings it back to original units
- makes it meaningful

Q10. What is correlation and what does it measure?

Answer:
Correlation measures the **strength and direction of linear relationship** between two variables.

Range:

- +1 → strong positive
- 0 → no relationship
- -1 → strong negative

Q11. Can correlation imply causation?

Answer:
No.

Example:
Ice cream sales and drowning both increase in summer.
They are correlated, but one does not cause the other.

Q12. What is the role of visualization in descriptive statistics?

Answer:
Visualization helps:

- detect patterns
- identify outliers
- understand distribution

Numbers alone cannot reveal structure clearly.

Q13. What is a histogram and what does it show?

Answer:
Histogram shows frequency distribution of data.

It answers:

- how data is distributed
- whether it is normal, skewed, or uniform

Q14. What is a boxplot and why is it useful?

Answer:
Boxplot shows:

- median
- quartiles
- outliers

Useful because:

- it quickly detects skewness and extreme values

Q15. Why is visualization necessary even after calculating statistics?

Answer:
Statistics summarize data but can hide patterns.

Example:
Two datasets can have:

- same mean
- same variance

But completely different distributions.

Visualization exposes this difference.

Reality Check

If you say:

- “Standard deviation measures spread”
- “Correlation shows relationship”

You will be cross-questioned immediately.

You must explain:

- why SD is better than range
- why correlation fails in nonlinear data
- how distribution affects mean vs median

PRACTICAL 4 — DATA ANALYTICS (BOSTON HOUSING DATASET)

(From your sheet)

Q1. What is the objective of the Boston Housing dataset problem?

Answer:
To predict **house prices (continuous value)** based on multiple features like crime rate, number of rooms, tax rate, etc.

This is a **regression problem**, not classification.

Q2. Why is this considered a regression problem?

Answer:
Because the target variable (price) is **continuous numeric**, not categorical.

If output is real-valued → regression
If output is category → classification

Q3. What are features and target variables in this dataset?

Answer:

- Features (independent variables): crime rate, rooms, age, etc.
- Target (dependent variable): house price

Features influence the target.

Q4. Why is data preprocessing important before applying regression?

Answer:
Because raw data may contain:

- missing values
- different scales
- irrelevant features

Regression assumes clean and properly scaled input. Otherwise, coefficients become unreliable.

Q5. What is multicollinearity?

Answer:
When two or more features are highly correlated with each other.

Example:

- Tax rate and property value may be correlated

Problem:

- Model cannot distinguish their individual effect
- Coefficients become unstable

Q6. What is Linear Regression?

Answer:
A model that assumes a linear relationship between input features and output.

It fits a line (or hyperplane in multiple dimensions) that minimizes error.

Q7. What does the regression equation represent?

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n$$

Answer:

- y→ predicted price
- β_0→ intercept
- β_i→ influence of each feature

Each coefficient shows how much that feature affects the output.

Q8. What is the meaning of coefficients in regression?

Answer:
They represent **impact of each feature** on the target.

Example:

- If $\beta_1 = 2$ → increasing feature by 1 increases price by 2 units

But interpretation is valid only if:

- features are independent
- scaling is handled

Q9. What assumptions does Linear Regression make?

Answer:

1. Linear relationship
2. No multicollinearity
3. Homoscedasticity (constant variance of errors)
4. Normal distribution of errors

Violation → model becomes unreliable

Q10. What happens if data is not linear?

Answer:
Model will **underfit**.

It cannot capture the pattern → predictions are poor.

Solution:

- Transform features
- Use nonlinear models

Q11. What is R^2 score?

Answer:
R^2 measures how much variance in target is explained by the model.

Range:

- 1 → perfect fit
- 0 → no explanatory power

Q12. Is high R^2 always good?

Answer:
No.

High R^2 can occur due to:

- overfitting
- too many features

Model may perform poorly on new data.

Q13. What is overfitting in this context?

Answer:
Model memorizes training data instead of learning pattern.

Result:

- high training accuracy
- low testing accuracy

Q14. Why do we split data into training and testing sets?

Answer:
To evaluate model on unseen data.

Without splitting:

- you test on training data
- results become misleading

Q15. What is the limitation of the Boston Housing dataset?

Answer:

- Small dataset
- Contains outdated and biased features
- Not suitable for real-world deployment

It is used only for learning, not production.

PRACTICAL 5 — CLASSIFICATION (SOCIAL NETWORK ADS DATASET)

(From your sheet)

Q1. What is the goal of the Social Network Ads dataset?

Answer:
To predict whether a user will **purchase a product (Yes/No)** based on features like age, salary, etc.

This is a **binary classification problem**.

Q2. Why is this not a regression problem?

Answer:
Because the output is categorical:

- 0 → Not Purchased
- 1 → Purchased

Regression predicts continuous values. Here we predict discrete classes.

Q3. What are independent and dependent variables here?

Answer:

- Independent (features): Age, Estimated Salary
- Dependent (target): Purchase (0 or 1)

Q4. Why is feature scaling important in this dataset?

Answer:
Because features have different ranges:

- Age: ~20–60
- Salary: thousands

Without scaling:

- algorithms like KNN, SVM, Logistic Regression become biased toward larger values

---

Q5. What is Logistic Regression and why is it used here?

Answer:
Logistic Regression is a classification algorithm that predicts probability of a class.

It is used because:

- output is binary
- it models probability between 0 and 1

Q6. What does the sigmoid function do?

$$P(y = 1 \mid x) = \frac{1}{1 + e^{-z}}$$

Answer:
It converts any real value into a probability between 0 and 1.

Q7. What is a decision boundary?

Answer:
A boundary that separates classes.

Example:

- If probability > 0.5 → class 1
- Else → class 0

It is usually a line (or curve) in feature space.

Q8. What is a confusion matrix?

Answer:
A table used to evaluate classification performance.

It contains:

- True Positive
- False Positive
- True Negative
- False Negative

Q9. Why is accuracy not always reliable?

Answer:
Because of class imbalance.

Example:

- 95% non-buyers

- Model predicts always “No” → 95% accuracy

But it fails to identify buyers.

Q10. What is precision?

Answer:
Precision measures how many predicted positives are actually correct.

High precision → low false positives

Q11. What is recall?

Answer:
Recall measures how many actual positives are correctly identified.

High recall → low false negatives

Q12. Difference between precision and recall?

Answer:

- Precision → correctness of predicted positives
- Recall → coverage of actual positives

Trade-off exists between them.

Q13. What is F1-score and why is it used?

Answer:
F1-score balances precision and recall.

Used when:

- both false positives and false negatives matter

Q14. What is overfitting in classification?

Answer:
Model performs very well on training data but poorly on unseen data.

Reason:

- memorization instead of generalization

Q15. Why do we use train-test split?

Answer:
To test model performance on unseen data.

Without it:

- model evaluation is biased
- results are misleading

Reality Check

If you say:

- “Accuracy is good metric”
- “Logistic regression is regression”

You won’t survive follow-ups.

You must explain:

- why accuracy fails
- when precision vs recall matters
- how decision boundary affects prediction

PRACTICAL 6 — NAIVE BAYES & DECISION TREE (IRIS DATASET)

(From your sheet)

Q1. What is the objective of this practical?

Answer:
To classify Iris flowers into species using:

- Naive Bayes
- Decision Tree

This is a **multi-class classification problem** (3 classes).

Q2. What type of dataset is the Iris dataset?

Answer:
A labeled dataset with:

- 4 numerical features
- 1 categorical target (species)

It is structured and clean, which is why it’s used for learning.

Q3. What is Naive Bayes?

Answer:
A probabilistic classifier based on Bayes theorem.

It calculates probability of a class given features and chooses the class with highest probability.

Q4. What is the “naive” assumption in Naive Bayes?

Answer:
It assumes all features are **independent** of each other.

This assumption is almost always false in real data, but the model still performs well.

Q5. Why does Naive Bayes work despite wrong assumptions?

Answer:
Because:

- it focuses on probability estimation

- independence assumption simplifies computation
- errors often cancel out

Q6. What is Decision Tree?

Answer:
A model that splits data into branches based on conditions.

Each node represents a decision rule.

Q7. How does a Decision Tree decide where to split?

Answer:
Using metrics like:

- Gini Index
- Entropy

It chooses the split that best separates classes.

Q8. What is entropy in Decision Trees?

$$Entropy = -\sum p_i \log_2(p_i)$$

Answer:
Measures impurity in data.

- High entropy → mixed classes
- Low entropy → pure classes

Q9. What is information gain?

Answer:
Reduction in entropy after a split.

Higher information gain → better split.

Q10. What is overfitting in Decision Trees?

Answer:
Tree becomes too complex and memorizes training data.

Result:

- perfect training accuracy
- poor test accuracy

Q11. How can overfitting be controlled in Decision Trees?

Answer:

- Limit tree depth
- Minimum samples per node
- Pruning

Q12. Difference between Naive Bayes and Decision Tree?

Answer:

Naive Bayes	**Decision Tree**
Probabilistic	Rule-based
Assumes independence	No such assumption
Fast	Can be complex
Works well with text	Works well with structured data

Q13. Which algorithm is better?

Answer:
Depends on data.

- Naive Bayes → high-dimensional data (text)
- Decision Tree → interpretable, nonlinear data

There is no universally best algorithm.

Q14. What is model interpretability?

Answer:
How easily humans can understand model decisions.

Decision Trees are highly interpretable.
Naive Bayes is less intuitive.

Q15. Why use two algorithms on the same dataset?

Answer:
To compare performance and understand:

- strengths
- weaknesses
- suitability for data

This helps in selecting the best model.

PRACTICAL 7 — CLUSTERING (K-MEANS ON IRIS DATASET)

Q1. What is clustering?

Answer:
Clustering is an **unsupervised learning technique** that groups similar data points together without using labeled outputs.

The goal is:

- maximize similarity within clusters

- minimize similarity between clusters

Q2. Why is K-Means called an unsupervised algorithm?

Answer:
Because it does not use target labels.

It only uses feature similarity to group data.

Q3. What is the role of K in K-Means?

Answer:
K represents the number of clusters.

You must define it before training.

Wrong K → meaningless clusters.

Q4. How does K-Means algorithm work?

Answer:

1. Initialize K centroids
2. Assign each point to nearest centroid
3. Recalculate centroids
4. Repeat until convergence

Q5. What is a centroid?

Answer:
The mean position of all points in a cluster.

It represents the “center” of that cluster.

Q6. What does K-Means try to minimize?

$$J = \sum_{i=1}^{k} \sum_{x \in C_i} \| x - \mu_i \|^2$$

Answer:
It minimizes the **sum of squared distances** between points and their cluster centroids.

Q7. Why is K-Means sensitive to initialization?

Answer:
Initial centroids are chosen randomly.

Different starting points → different final clusters.

This can lead to inconsistent results.

Q8. What is the Elbow Method?

Answer:
A technique to choose optimal K.

You plot:

- K vs error (inertia)

The point where improvement slows down is chosen.

Q9. Why is choosing K difficult?

Answer:
Because:

- no ground truth in unsupervised learning
- elbow is not always clear
- different K values may seem valid

Q10. What are limitations of K-Means?

Answer:

- assumes spherical clusters
- sensitive to outliers
- requires predefined K
- fails with non-linear clusters

Q11. Why does K-Means fail with non-spherical data?

Answer:
Because it uses distance from centroid.

If clusters are irregular (e.g., spiral shape), distance-based grouping fails.

Q12. What is inertia in K-Means?

Answer:
Inertia is the total within-cluster distance.

Lower inertia → tighter clusters.

Q13. Can clustering results be validated without labels?

Answer:
Yes, using internal metrics:

- inertia
- silhouette score

But evaluation is still approximate.

Q14. Difference between clustering and classification?

Answer:

Clustering	Classification
No labels	Uses labels

Clustering	Classification
Finds structure	Predicts categories
Exploratory	Predictive

Q15. Why apply K-Means on Iris dataset if labels already exist?

Answer:
To test whether clustering can discover natural groupings.

You compare clusters with actual labels to evaluate performance.

Reality Check

If you say:

- "K-Means groups similar data"
- "Elbow method finds best K"

Practical 8: Data Visualization (Seaborn + Matplotlib on dataset)
This one exposes whether you actually understand what your plots mean or you just draw them.

next

If your answer is "we used seaborn to plot graphs," you'll get shut down. This practical is about **why a specific visualization is chosen and what insight it reveals**.

PRACTICAL 8 — DATA VISUALIZATION (SEABORN & MATPLOTLIB)

(From your sheet)

Q1. What is the purpose of data visualization in this practical?

Answer:
To convert numerical data into visual form so patterns, relationships, and anomalies become obvious.

It is not for decoration. It is for **interpretation and decision-making**.

Q2. Why do we use visualization even after statistical analysis?

Answer:
Statistics summarize data but can hide structure.

Two datasets can have:

- same mean
- same standard deviation

But completely different distributions.

Visualization exposes those differences.

Q3. Difference between Matplotlib and Seaborn?

Answer:

- Matplotlib → low-level, more control
- Seaborn → high-level, better aesthetics, built on Matplotlib

Seaborn simplifies statistical plotting.

Q4. When should you use a bar chart?

Answer:
When comparing **categorical data**.

Example:

- average marks per subject

Using it for continuous distribution is wrong.

Q5. When should you use a histogram?

Answer:
To show distribution of a single continuous variable.

It helps identify:

- skewness
- spread
- outliers

Q6. What does a scatter plot represent?

Answer:
Relationship between two variables.

It shows:

- correlation
- clusters
- trends

Q7. What is a boxplot and why is it powerful?

Answer:
A boxplot shows:

- median
- quartiles
- outliers

It quickly reveals:

- spread
- skewness
- extreme values

Q8. What is overplotting and how do you handle it?

Answer:
When too many points overlap, making visualization unclear.

Solutions:

- reduce opacity (alpha)
- sampling
- use hexbin plots

Q9. What is the importance of labeling in graphs?

Answer:
Without labels:

- graph is meaningless

Must include:

- title
- axis labels
- legend

Q10. What are common mistakes in visualization?

Answer:

- wrong chart type
- misleading axis scale
- too many colors
- cluttered design

These lead to wrong interpretation.

Q11. Why is a pie chart often discouraged?

Answer:
Because humans cannot compare angles accurately.

Bar charts are more precise for comparison.

Q12. What is correlation heatmap?

Answer:
A matrix showing correlation between variables using colors.

Helps identify:

- strong relationships
- redundant features

Q13. How does visualization help in feature selection?

Answer:
By showing:

- correlation between features

- relationship with target

Highly correlated features may be removed.

Q14. What happens if visualization is misleading?

Answer:

- Wrong conclusions
- Poor decisions
- Invalid analysis

Visualization can distort truth if designed incorrectly.

Q15. Why is visualization important for non-technical stakeholders?

Answer:
Because they cannot interpret raw data or models.

Visualization:

- simplifies complex information
- supports decision-making

Reality Check

If you say:

- "Histogram shows data"
- "Scatter shows relationship"

You'll be pushed further.

You must explain:

- what specific insight each graph gives
- why that graph is chosen over others
- how misinterpretation happens

You've covered all practicals.

If you want next:

- I can generate **FULL external viva (50–100 killer questions mixed across all units)**
- Or simulate a **real viva where I cross-question you until you break**

next

Good. Now no more isolated practicals. This is where most students collapse.

I'm going to give you a **full external viva set**. Mixed questions. No pattern. Exactly how examiners hit you.

PRACTICAL — TITANIC DATASET (EDA + VISUALIZATION)

Q1. What is the purpose of loading the Titanic dataset?

Answer:
To perform **Exploratory Data Analysis (EDA)**.

EDA helps:

- understand structure
- detect missing values
- identify patterns (like survival trends)

This step happens before modeling.

Q2. What does sns.load_dataset("titanic") do?

Answer:
It loads a built-in dataset from Seaborn.

- No need for CSV file
- Already cleaned to some extent
- Used for learning and visualization

Q3. What information does df.head() provide?

Answer:
It shows first 5 rows of dataset.

Used to:

- understand columns
- preview data values
- detect obvious issues

Q4. What does df.shape tell you?

Answer:
It returns:

(rows, columns)

This tells:

- dataset size
- number of features

Q5. Why do we check df.isnull().sum()?

Answer:
To detect missing values in each column.

Missing values:

- affect analysis
- break models

- need handling

Q6. Which columns in Titanic dataset typically have missing values?

Answer:

- age
- deck
- embark_town

Among these:

- age is important → should be handled carefully
- deck often dropped due to too many missing values

Q7. What is a boxplot and why is it used here?

Answer:
Boxplot shows:

- median
- quartiles
- outliers

Used here to:

- compare age distribution
- across gender and survival

Q8. What does x="sex" represent in the plot?

Answer:
It divides data into categories:

- male
- female

So comparison is done based on gender.

Q9. What does y="age" represent?

Answer:
It represents the numerical variable being analyzed.

Here:

- age distribution is studied

Q10. What is the role of hue="survived"?

Answer:
It adds an additional grouping:

- 0 → did not survive

- 1 → survived

This allows comparison:

- within gender
- across survival status

Q11. What insight do you get about gender and survival?

Answer:
Female passengers had significantly higher survival rates.

Reason (real-world context):

- evacuation priority given to women

Q12. What does the presence of outliers in boxplot indicate?

Answer:
Outliers are extreme values.

In this case:

- very high age passengers

They may be:

- valid data
- or rare cases

Q13. What does median line in boxplot represent?

Answer:
It represents the **middle value** of the data.

It helps compare:

- central tendency across groups

Q14. Why is visualization better than raw numbers here?

Answer:
Because:

- patterns are visible instantly
- group comparisons are easier
- outliers are clearly shown

Raw numbers cannot show relationships effectively.

Q15. What conclusions can you draw from this visualization?

Answer:

1. Females had higher survival rate
2. Children had better survival chances
3. Male non-survivors dominate

4. Age distribution varies across groups

5. Outliers exist in higher age

Q16. What are limitations of this analysis?

Answer:

- Only age and gender considered
- ignores other factors (class, fare, etc.)
- visualization does not prove causation

Q17. What preprocessing step is needed before modeling this dataset?

Answer:

- handle missing values
- encode categorical variables
- remove irrelevant columns

Q18. Why is plt.figure(figsize=(10,6)) used?

Answer:
To control size of the plot.

Larger size:

- improves readability
- avoids overlapping elements

Q19. What happens if hue is removed?

Answer:
You lose survival comparison.

Plot will only show:

- age vs gender

No survival insight.

Q20. Can this visualization be misleading?

Answer:
Yes.

Reasons:

- sample imbalance
- missing data
- ignoring other variables

Visualization shows pattern, not complete truth.

Reality Check

If you say:

- "Boxplot shows distribution"
- "Females survived more"

You'll get destroyed.

You must explain:

- how boxplot represents data mathematically
- why that pattern appears
- what limitations exist

PRACTICAL — IRIS DATASET (FEATURE DISTRIBUTION & HISTOGRAM ANALYSIS)

Q1. What is the objective of this practical?

Answer:
To analyze **distribution of features** in the Iris dataset using histograms and identify:

- spread
- skewness
- outliers
- feature separability

This helps decide which features are useful for classification.

Q2. What type of dataset is the Iris dataset?

Answer:
A structured, labeled dataset with:

- 4 numerical features
- 1 categorical target (species)

It is used for **multi-class classification**.

Q3. What does df.dtypes reveal and why is it important?

Answer:
It shows data types of each column.

Importance:

- ensures all features are numeric
- confirms compatibility with ML algorithms

If types are wrong, models fail or behave incorrectly.

Q4. Why are only four features selected?

Answer:
Because these are the actual measurable attributes:

- sepal length
- sepal width
- petal length
- petal width

The target (species) is excluded because it is not used in histogram plotting.

Q5. What is a histogram?

Answer:
A histogram shows **frequency distribution** of a variable.

It divides data into bins and counts how many values fall into each bin.

Q6. Why use histograms instead of bar charts here?

Answer:

- Histogram → continuous data
- Bar chart → categorical data

Using bar chart here would be incorrect because features are continuous.

Q7. What does bins=15 represent?

Answer:
It divides data into 15 intervals.

More bins:

- more detail
- more noise

Fewer bins:

- smoother but less precise

Q8. What does the shape of histogram tell you?

Answer:
It reveals:

- distribution type (normal, skewed)
- clustering patterns
- presence of outliers

Q9. Why do petal length and width show clear grouping?

Answer:
Because different species have distinct petal sizes.

This makes:

- clusters visible
- classification easier

Q10. Why are sepal features less effective for classification?

Answer:
Because their distributions overlap across species.

Overlap → difficult to separate classes.

Q11. What does “moderate spread” in sepal length mean?

Answer:
Values are distributed across a range without extreme clustering.

It is neither tightly grouped nor highly dispersed.

Q12. How do you detect outliers in histogram?

Answer:
Outliers appear as:

- isolated bars far from main distribution

However, histograms are less precise than boxplots for outlier detection.

Q13. Why is sepal width said to contain outliers?

Answer:
Because some values fall far outside the main cluster.

This indicates:

- unusual measurements
- or natural variation

Q14. Why are boxplots better for detecting outliers than histograms?

Answer:
Because boxplots explicitly show:

- quartiles
- whiskers
- outliers

Histograms only approximate distribution.

Q15. What is feature separability and why is it important?

Answer:
Feature separability means how well a feature distinguishes between classes.

High separability:

- better model performance

Low separability:

- confusion between classes

Q16. What would happen if all features had overlapping distributions?

Answer:
Model would struggle to classify data.

Result:

- low accuracy
- poor decision boundaries

Q17. Why is visualization done before modeling?

Answer:
To:

- understand data behavior
- select useful features
- detect issues

Without this, model design is blind.

Q18. What is skewness and how is it visible here?

Answer:
Skewness indicates asymmetry in distribution.

- Right skew → tail on right
- Left skew → tail on left

Histogram shape reveals this.

Q19. Why is frequency important in histogram?

Answer:
It shows how often values occur.

High frequency:

- common values

Low frequency:

- rare values or outliers

Q20. What is the limitation of analyzing features individually?

Answer:
It ignores relationships between features.

Example:

- two features may be weak individually
- but strong together

So pairwise or multivariate analysis is needed.

FULL DATA SCIENCE VIVA — HIGH PRESSURE SET

Q1. Why is data preprocessing more important than model selection?

Answer:
Because models learn patterns from data. If data is:

- noisy
- biased
- inconsistent

Then even the best algorithm produces wrong results.

Garbage in → garbage out.

Q2. You have missing values in 40% of a column. What will you do?

Answer:
You don't blindly fill.

Options:

- If feature is important → use advanced imputation
- If irrelevant → drop column
- If dataset is small → careful imputation

Decision depends on:

- importance of feature
- impact on model

Q3. Why is scaling required for K-Means but not for Decision Trees?

Answer:

- K-Means uses **distance calculations** → scale matters
- Decision Trees use **threshold splits** → scale irrelevant

Q4. Why does linear regression fail on nonlinear data?

Answer:
Because it assumes linear relationship.

If data is curved:

- model underfits
- error remains high

Q5. You get 99% accuracy. Why might your model still be useless?

Answer:
Because of class imbalance.

Example:

- 99% negative class
- model predicts only negative

Accuracy looks high, but model is useless.

Q6. What is overfitting and how do you detect it?

Answer:
Overfitting = model memorizes training data.

Detection:

- high training accuracy
- low testing accuracy

Q7. Difference between Type I and Type II error?

Answer:

- Type I → reject true hypothesis
- Type II → accept false hypothesis

Q8. Why is correlation not equal to causation?

Answer:
Because two variables can move together due to a third hidden factor.

Correlation only shows relationship, not cause-effect.

Q9. Why does Naive Bayes work even when independence assumption is false?

Answer:
Because:

- it estimates probabilities efficiently
- simplification reduces complexity
- practical performance remains strong

Q10. What happens if you choose wrong value of K in K-Means?

Answer:

- Too small K → different groups merged
- Too large K → same group split unnecessarily

Both lead to meaningless clusters.

Q11. Why is median preferred for skewed data?

Answer:
Median is not affected by extreme values.

Mean gets distorted.

Q12. What is multicollinearity and why is it harmful?

Answer:
When features are highly correlated.

Problem:

- unstable coefficients
- unreliable interpretation

Q13. Why do we split data into training and testing sets?

Answer:
To evaluate model on unseen data.

Without split:

- evaluation becomes biased

Q14. What is feature engineering?

Answer:
Creating new features from existing data to improve model performance.

Example:
Date → extract month, day

Q15. Why is visualization necessary before modeling?

Answer:
To understand:

- distribution
- outliers
- relationships

Without this, model decisions are blind.

Q16. Why is standard deviation better than range?

Answer:
Range uses only min and max.

Standard deviation considers all data points.

Q17. What is decision boundary in classification?

Answer:
A line or surface that separates classes in feature space.

Q18. Why do decision trees overfit easily?

Answer:
Because they keep splitting until data is perfectly classified.

This leads to memorization.

Q19. What is the difference between supervised and unsupervised learning?

Answer:

- Supervised → uses labeled data
- Unsupervised → no labels

Q20. Why is F1-score better than accuracy in some cases?

Answer:
Because it balances precision and recall.

Used when:

- both false positives and false negatives matter

Q21. Why is Hadoop needed? Why not use a single machine?

Answer:
Single machine:

- limited storage
- limited processing

Hadoop:

- distributed storage
- parallel processing

Q22. What is HDFS replication and why is it important?

Answer:
Data is stored in multiple copies across nodes.

If one node fails:

- data is not lost

Q23. Why is MapReduce considered slow today?

Answer:
Because:

- writes intermediate data to disk
- multiple stages increase latency

Modern systems use in-memory processing.

Q24. What is bias vs variance tradeoff?

Answer:

- High bias → underfitting
- High variance → overfitting

Goal:
Balance both.

Q25. Why is model interpretability important?

Answer:
Because decisions must be explainable, especially in:

- healthcare
- finance

Final Reality Check

If you answer like:

- “Overfitting means model is too complex”
- “Scaling improves performance”

You’ll be pushed until you fail.

You must explain:

- mechanism behind each concept
- when it applies
- what happens if ignored